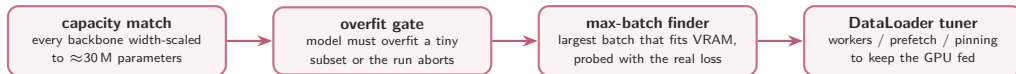


DLR-TomoSAR — Benchmark & winner

run-health tooling · every backbone × every objective · winner under the microscope ·
dual trunks · paired losses

July 2026

Run-health tooling: no GPU-hour starts unhealthy



7 backbones unet, unet_skip, resnet, attention_unet, deeplabv3plus, segformer, nafnet

5 objectives curve space: MSE, L1, Charbonnier · parameter space: MSE, L1

1 head, fixed gated set prediction · Hungarian matching — the Act III verdict, baked into every cell

5 seeds every cell trained five times — 35 configurations $\times$ 5 seeds = 175 runs, reported mean $\pm$ seed std

- Capacity is matched, the recipe is identical in every cell (set-pred · Hungarian · presence lever A · hv flips · $K=2$; reference cell **UNet + param-MSE**) — differences are architectural, not capacity.
- A model that cannot memorise a handful of patches has a plumbing, loss or architecture bug — caught in minutes, not after hours. Surrogate-loss memory probes understate the footprint $\approx 10\times$, so the batch is probed with the genuine training step and the LR auto-scales with it.
- Payoff: the benchmark grid runs unattended — a dead cell means a real model failure, not an infrastructure one.

The board at five seeds — the L1 family owns the top nine

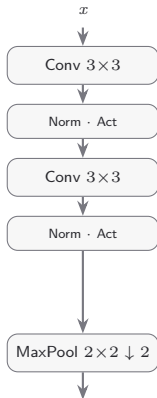
#	model	loss	curve $R^2 \uparrow$	cosine $\uparrow$	peak err $\downarrow$	recall $\uparrow$	score
1	unet-skip	L1-curve	0.756 $\pm .012$	0.924 $\pm .007$	1.29 $\pm .05$	0.886 $\pm .006$	0.983
2	unet-skip	Charbonnier	0.751 $\pm .012$	0.928 $\pm .004$	1.25 $\pm .03$	0.886 $\pm .004$	0.977
3	unet-skip	param-L1	0.741 $\pm .016$	0.931 $\pm .001$	1.38 $\pm .03$	0.893 $\pm .004$	0.963
4	resunet	param-L1	0.734 $\pm .013$	0.927 $\pm .007$	1.32 $\pm .07$	0.894 $\pm .003$	0.946
5	deeplab	param-L1	0.731 $\pm .006$	0.923 $\pm .004$	1.35 $\pm .05$	0.888 $\pm .002$	0.937
6	deeplab	L1-curve	0.731 $\pm .012$	0.921 $\pm .013$	1.33 $\pm .07$	0.873 $\pm .022$	0.935
7	resunet	L1-curve	0.737 $\pm .019$	0.916 $\pm .007$	1.34 $\pm .06$	0.881 $\pm .005$	0.930
8	resunet	Charbonnier	0.731 $\pm .016$	0.921 $\pm .014$	1.31 $\pm .08$	0.881 $\pm .006$	0.929
9	deeplab	Charbonnier	0.735 $\pm .008$	0.913 $\pm .015$	1.62 $\pm .45$	0.868 $\pm .037$	0.912
10	nafnet	Charbonnier	0.702 $\pm .009$	0.906 $\pm .007$	1.48 $\pm .05$	0.875 $\pm .008$	0.829
11	segformer	Charbonnier	0.688 $\pm .022$	0.910 $\pm .016$	1.51 $\pm .20$	0.866 $\pm .019$	0.829
12	nafnet	L1-curve	0.698 $\pm .007$	0.908 $\pm .010$	1.43 $\pm .08$	0.879 $\pm .005$	0.828
⋮							
16	deeplab	MSE-curve	0.732 $\pm .010$	0.872 $\pm .019$	1.62 $\pm .08$	0.869 $\pm .004$	0.785
20	unet	L1-curve	0.660 $\pm .015$	0.897 $\pm .026$	1.51 $\pm .28$	0.870 $\pm .029$	0.760
21	unet-skip	param-MSE	0.647 $\pm .047$	0.879 $\pm .017$	1.59 $\pm .13$	0.881 $\pm .022$	0.692

runs: model · set-pred · Hungarian · $K=2$ · hv flips · presence A · loss · patch 64×32 , five seeds; held-out band. Cells: mean $\pm$ seed std; **bold** = best mean in column (rows shown), marked only where best and worst differ by more than 5%. score = equal-weight mean of the five metric groups (curve error, variance, SSIM, cosine, peak), min-max scaled over the 35 arms; # = rank; ranks 13–15, 17–19, 22–35 on the next frame only. cosine = per-pixel profile cosine mean; peak err in elevation units; recall = matched. **green** = the winner's score; **red** = the param-MSE reference cell, 21st of 35.

One encoder block: UNet vs UNet-skip vs ResUNet

UNet

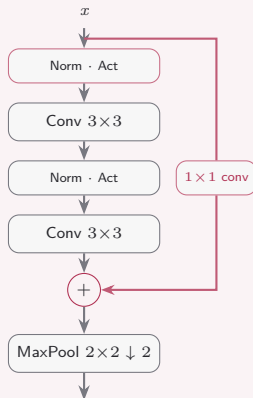
board, L1-curve: 0.760



plain double conv, no shortcut;
downsampling by max-pool

UNet-skip

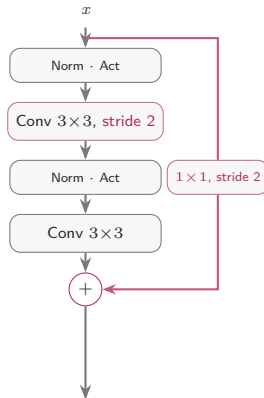
board, L1-curve: 0.983



pre-activation + **residual shortcut**;
max-pool kept

ResUNet

board, L1-curve: 0.930



downsampling inside the block: **stride-2**
conv and shortcut, no pooling (stage 1: stride 1)

The board by scatterer count — curve-MSE drops the single scatterer

metric	UNet-skip — winner backbone					UNet
	L1	Charb.	param-L1	MSE	param-MSE	param-MSE
matched precision $\uparrow$	0.893 $\pm$.005	0.896 $\pm$.006	0.900 $\pm$.001	0.858 $\pm$.012	0.816 $\pm$.041	0.820 $\pm$.006
$k=1$	0.916 $\pm$.006	0.922 $\pm$.008	0.948 $\pm$.002	0.883 $\pm$.016	0.830 $\pm$.059	0.843 $\pm$.008
$k=2$	0.850 $\pm$.005	0.850 $\pm$.004	0.821 $\pm$.002	0.821 $\pm$.006	0.792 $\pm$.011	0.779 $\pm$.006
matched recall $\uparrow$	0.886 $\pm$.006	0.886 $\pm$.004	0.893 $\pm$.004	0.766 $\pm$.072	0.881 $\pm$.022	0.894 $\pm$.002
$k=1$	0.997 $\pm$.002	0.998 $\pm$.001	0.997 $\pm$.001	0.808 $\pm$.122	0.981 $\pm$.030	0.997 $\pm$.001
$k=2$	0.728 $\pm$.013	0.727 $\pm$.008	0.745 $\pm$.008	0.707 $\pm$.013	0.739 $\pm$.014	0.748 $\pm$.003
count acc pred $k=1 \uparrow$	0.910 $\pm$.006	0.906 $\pm$.006	0.950 $\pm$.003	0.922 $\pm$.014	0.973 $\pm$.008	0.973 $\pm$.004
count acc pred $k=2 \uparrow$	0.740 $\pm$.011	0.753 $\pm$.015	0.847 $\pm$.005	0.723 $\pm$.010	0.621 $\pm$.083	0.640 $\pm$.013
count exact $\uparrow$	0.861 $\pm$.002	0.865 $\pm$.002	0.917 $\pm$.004	0.723 $\pm$.088	0.812 $\pm$.082	0.845 $\pm$.008
matched μ MAE $\downarrow$	1.49 $\pm$.04	1.48 $\pm$.03	1.72 $\pm$.04	2.01 $\pm$.16	1.94 $\pm$.09	2.19 $\pm$.17
$k=1$	0.39 $\pm$.04	0.37 $\pm$.02	0.34 $\pm$.01	0.68 $\pm$.11	0.50 $\pm$.05	0.71 $\pm$.23
$k=2$	3.32 $\pm$.06	3.33 $\pm$.03	3.87 $\pm$.07	3.76 $\pm$.08	4.11 $\pm$.15	4.38 $\pm$.13
matched σ MAE $\downarrow$	0.77 $\pm$.04	0.78 $\pm$.01	0.73 $\pm$.01	1.17 $\pm$.07	0.72 $\pm$.02	0.81 $\pm$.01
$k=1$	0.22 $\pm$.01	0.23 $\pm$.02	0.20 $\pm$.00	0.39 $\pm$.02	0.21 $\pm$.02	0.24 $\pm$.01
$k=2$	1.67 $\pm$.08	1.70 $\pm$.05	1.57 $\pm$.03	2.21 $\pm$.04	1.48 $\pm$.03	1.66 $\pm$.03
matched a MAE $\downarrow$	0.241 $\pm$.011	0.242 $\pm$.008	0.175 $\pm$.002	0.387 $\pm$.030	0.223 $\pm$.010	0.307 $\pm$.011
$k=1$	0.090 $\pm$.008	0.090 $\pm$.011	0.055 $\pm$.001	0.176 $\pm$.030	0.089 $\pm$.012	0.125 $\pm$.011
$k=2$	0.491 $\pm$.024	0.494 $\pm$.013	0.363 $\pm$.007	0.669 $\pm$.020	0.424 $\pm$.011	0.576 $\pm$.017

runs: UNet-skip / UNet · set-pred · Hungarian · $K=2$ · hv flips · presence A · patch 64×32 (backbone · objective = column); held-out validation band, cells mean $\pm$ seed std over five seeded replicas; k = true scatterer count of the pixel. **bold** = best in row, marked only where best and worst differ by more than 5%. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude. **green** = param-L1's count and amplitude calibration; **red** = curve-MSE dropping the single scatterer.

The board, predicted statistics vs the GT — the gate sets activity, param-L1 calibrates amplitude

prediction mean	GT	UNet-skip — winner backbone					UNet
		L1	Charb.	param-L1	MSE	param-MSE	param-MSE
slot 0 — strongest scatterer; GT-active on every pixel							
active frac	1.00	0.97 ±.01	0.97 ±.00	0.98 ±.00	0.82 ±.09	0.97 ±.02	0.99 ±.00
a — active	0.77	0.66 ±.01	0.66 ±.00	0.74 ±.00	0.65 ±.06	0.72 ±.02	0.59 ±.02
μ — active	-4.89	-4.50 ±.12	-4.56 ±.11	-4.52 ±.04	-4.02 ±.24	-4.19 ±.09	-4.08 ±.21
σ — active	1.76	1.39 ±.05	1.37 ±.06	1.38 ±.01	1.59 ±.13	1.41 ±.06	1.38 ±.05
slot 1 — second scatterer; GT-active on 26% of pixels							
active frac	0.260	0.280 ±.010	0.274 ±.011	0.274 ±.003	0.301 ±.007	0.392 ±.059	0.384 ±.009
a — active	1.43	1.25 ±.07	1.24 ±.05	1.42 ±.02	1.05 ±.05	1.32 ±.05	0.95 ±.02
a — inact.	0.00	0.69 ±.02	0.71 ±.07	0.88 ±.06	0.54 ±.04	0.41 ±.13	0.34 ±.05
μ — active	11.34	6.11 ±.26	5.99 ±.19	8.30 ±.25	6.41 ±.16	8.77 ±.40	8.18 ±.39
μ — inact.	—	0.35 ±.73	0.09 ±.26	0.83 ±.30	1.35 ±.40	-0.51 ±1.26	-0.17 ±.62
σ — active	3.53	3.98 ±.15	3.93 ±.20	3.22 ±.03	5.57 ±.16	3.10 ±.13	2.77 ±.05
σ — inact.	—	1.75 ±.10	1.79 ±.06	1.81 ±.04	3.42 ±.14	1.56 ±.21	1.64 ±.08
distribution width — std over GT-active pixels							
a — slot 0	1.48	1.23 ±.07	1.24 ±.02	1.36 ±.02	1.18 ±.06	1.35 ±.08	0.92 ±.02
μ — slot 0	3.30	1.98 ±.07	1.97 ±.08	2.56 ±.06	1.80 ±.16	2.62 ±.11	2.14 ±.13
σ — slot 0	2.84	1.10 ±.05	1.05 ±.08	1.23 ±.03	1.08 ±.05	1.18 ±.06	1.03 ±.05
a — slot 1	1.45	1.17 ±.07	1.16 ±.04	1.29 ±.01	1.01 ±.04	1.23 ±.04	0.83 ±.03
μ — slot 1	16.24	7.92 ±.42	7.93 ±.39	9.98 ±.31	7.15 ±.35	9.94 ±.49	8.88 ±.48
σ — slot 1	4.01	2.08 ±.14	2.14 ±.39	1.77 ±.04	2.68 ±.13	1.56 ±.06	1.17 ±.06

runs: UNet-skip / UNet · set-pred · Hungarian · $K=2$ · hv flips · presence A · patch 64×32 (backbone · objective = column); five seeded replicas per arm, cells mean ± seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ, σ in elevation units; *a* in linear amplitude. **green** = param-L1's amplitude calibration; **red** = curve-MSE's gate collapse.

Every headline metric, all 35 configurations

#	model	loss	R^2	PSNR	cosine	SSIM elev	SSIM range	SSIM azim	F1	recall	2-scatterer recall	precision	μ MAE	σ MAE	a MAE	peak error	score
1	unet-skip	L1-curve	0.76	52.8	0.924	0.968	0.950	0.961	0.889	0.886	0.728	0.893	1.49	0.77	0.241	1.29	0.983
2	unet-skip	Charbonnier	0.75	52.7	0.928	0.967	0.950	0.961	0.891	0.886	0.727	0.896	1.48	0.78	0.242	1.25	0.977
3	unet-skip	param-L1	0.74	52.5	0.931	0.967	0.951	0.961	0.897	0.893	0.745	0.900	1.72	0.73	0.175	1.38	0.963
4	resunet	param-L1	0.73	52.4	0.927	0.966	0.950	0.960	0.898	0.894	0.746	0.902	1.75	0.76	0.178	1.32	0.946
5	deeplab	param-L1	0.73	52.4	0.923	0.966	0.949	0.960	0.895	0.888	0.736	0.902	1.71	0.72	0.184	1.35	0.937
6	deeplab	L1-curve	0.73	52.4	0.921	0.967	0.949	0.960	0.876	0.873	0.721	0.879	1.52	0.74	0.241	1.33	0.935
7	resunet	L1-curve	0.74	52.5	0.916	0.967	0.949	0.960	0.890	0.881	0.715	0.900	1.46	0.77	0.238	1.34	0.930
8	resunet	Charbonnier	0.73	52.4	0.921	0.967	0.949	0.960	0.889	0.881	0.716	0.897	1.49	0.75	0.236	1.31	0.929
9	deeplab	Charbonnier	0.73	52.4	0.913	0.967	0.949	0.960	0.858	0.868	0.722	0.851	1.55	0.74	0.244	1.62	0.912
10	nafnet	Charbonnier	0.70	51.9	0.906	0.965	0.944	0.957	0.878	0.875	0.711	0.881	1.56	0.76	0.261	1.48	0.829
11	segformer	Charbonnier	0.69	51.7	0.910	0.965	0.945	0.957	0.873	0.866	0.704	0.881	1.54	0.76	0.265	1.51	0.829
12	nafnet	L1-curve	0.70	51.9	0.908	0.965	0.944	0.957	0.880	0.879	0.714	0.881	1.56	0.78	0.264	1.43	0.828
13	segformer	param-L1	0.67	51.5	0.919	0.964	0.945	0.956	0.883	0.884	0.728	0.883	1.76	0.73	0.196	1.39	0.811
14	segformer	L1-curve	0.68	51.6	0.905	0.965	0.943	0.956	0.870	0.864	0.703	0.876	1.59	0.77	0.274	1.56	0.804
15	nafnet	param-L1	0.67	51.5	0.917	0.963	0.945	0.956	0.885	0.883	0.724	0.888	1.77	0.75	0.198	1.43	0.797
16	deeplab	MSE-curve	0.73	52.4	0.872	0.962	0.937	0.953	0.861	0.869	0.710	0.853	1.84	1.07	0.334	1.62	0.785
17	att-unet	L1-curve	0.67	51.4	0.899	0.965	0.945	0.957	0.863	0.885	0.725	0.843	1.77	0.83	0.312	1.44	0.772
18	unet	Charbonnier	0.66	51.4	0.901	0.965	0.945	0.957	0.866	0.880	0.713	0.854	1.69	0.79	0.324	1.45	0.772
19	att-unet	Charbonnier	0.67	51.4	0.890	0.965	0.944	0.957	0.862	0.884	0.723	0.842	1.79	0.81	0.307	1.47	0.764
20	unet	L1-curve	0.66	51.4	0.897	0.965	0.944	0.956	0.863	0.870	0.711	0.858	1.71	0.81	0.328	1.51	0.760
21	unet-skip	param-MSE	0.65	51.2	0.879	0.963	0.940	0.954	0.847	0.881	0.739	0.816	1.94	0.72	0.223	1.59	0.692
22	resunet	param-MSE	0.64	51.1	0.874	0.961	0.938	0.952	0.851	0.880	0.730	0.824	1.95	0.73	0.235	1.60	0.653
23	unet-skip	MSE-curve	0.73	52.3	0.767	0.963	0.932	0.953	0.809	0.766	0.707	0.858	2.01	1.17	0.387	2.66	0.614
24	deeplab	param-MSE	0.60	50.7	0.883	0.960	0.936	0.950	0.841	0.855	0.716	0.828	1.94	0.73	0.243	1.73	0.613
25	unet	param-L1	0.61	50.8	0.855	0.963	0.940	0.953	0.893	0.886	0.725	0.900	2.01	0.80	0.251	1.56	0.610
26	att-unet	param-L1	0.61	50.7	0.885	0.963	0.942	0.954	0.891	0.889	0.732	0.894	2.06	0.82	0.255	1.49	0.588
27	resunet	MSE-curve	0.72	52.3	0.756	0.963	0.931	0.953	0.775	0.713	0.695	0.851	2.04	1.19	0.404	3.18	0.573
28	segformer	param-MSE	0.57	50.4	0.868	0.959	0.934	0.947	0.832	0.871	0.702	0.796	2.03	0.75	0.243	1.68	0.532
29	unet	param-MSE	0.58	50.4	0.829	0.960	0.934	0.949	0.856	0.894	0.748	0.820	2.19	0.81	0.307	1.67	0.484
30	att-unet	param-MSE	0.58	50.4	0.853	0.960	0.934	0.949	0.853	0.892	0.742	0.817	2.16	0.81	0.311	1.66	0.477
31	nafnet	param-MSE	0.58	50.4	0.877	0.958	0.932	0.946	0.837	0.871	0.709	0.807	2.09	0.79	0.260	1.69	0.437
32	segformer	MSE-curve	0.67	51.5	0.712	0.956	0.922	0.945	0.741	0.676	0.676	0.823	2.17	1.45	0.448	3.89	0.346
33	unet	MSE-curve	0.65	51.2	0.760	0.953	0.914	0.939	0.797	0.765	0.670	0.834	2.17	1.55	0.478	2.78	0.342
34	att-unet	MSE-curve	0.63	51.0	0.759	0.951	0.911	0.936	0.818	0.802	0.680	0.837	2.21	1.66	0.466	2.47	0.307
35	nafnet	MSE-curve	0.69	51.7	0.680	0.958	0.921	0.946	0.730	0.663	0.687	0.821	2.15	1.54	0.426	4.44	0.279

Five-seed means vs GT (spreads on the previous frame and in the per-arm reports) — PSNR in dB; cosine = per-pixel profile cosine mean; F1 / recall / precision matched, 2-scatterer recall on GT $k=2$ pixels; μ/σ MAE and peak error in elevation units, a MAE in linear amplitude; score = five-group composite (row order). **Bold** = best in column, marked only where best and worst differ by more than 5%; grey rows = MSE-family objectives. runs: set-pred · Hungarian · $K=2$ · hv flips · presence A · patch 64×32 .

Where the gain comes from: the head, not the matching

loss	conv sorted	conv Hungarian	set-pred sorted	set-pred Hungarian	Δ total	Δ head	Δ matching
<i>ResUNet</i>							
MSE-curve	0.608	—	0.926	0.926	+0.318	+0.318	0.000
Huber-curve	0.670	—	0.945	0.945	+0.275	+0.275	0.000
Charbonnier	0.914	—	0.991	0.991	+0.077	+0.077	0.000
param-MSE	0.937	0.924	—	0.939	+0.002	+0.016	−0.013
param-Huber	0.961	0.961	—	0.972	+0.011	+0.011	0.000
param-L1	0.982	0.982	0.992	0.988	+0.006	+0.006	−0.001
<i>Attention-UNet</i>							
MSE-curve	0.556	—	0.807	0.671	+0.115	+0.251	−0.136
Huber-curve	0.624	—	0.929	0.922	+0.298	+0.305	−0.007
Charbonnier	0.790	—	0.960	0.962	+0.172	+0.170	+0.003
param-MSE	0.855	0.849	—	0.892	+0.037	+0.044	−0.006
param-Huber	0.853	0.849	—	0.913	+0.060	+0.064	−0.003
param-L1	0.936	0.921	0.942	0.942	+0.006	+0.021	−0.015
<i>DeepLabV3+</i> (total only — 2×2 not completed)							
MSE-curve	0.818	—	—	0.930	+0.112	—	—
Huber-curve	0.831	—	—	0.964	+0.132	—	—
Charbonnier	0.641	—	—	0.964	+0.323	—	—
param-MSE	0.856	—	—	0.934	+0.078	—	—
param-Huber	0.971	—	—	0.984	+0.013	—	—
param-L1	0.983	—	—	0.992	+0.008	—	—

composite score per cell; — = not run. Δ_{total} = set-pred-Hungarian − conv-sorted, split via the completed corner into Δ_{head} + Δ_{matching} . **green** = the head effect on curve losses — the matching column never leaves zero.

Hungarian at $K=2$: costs nothing — and that is the point

model	loss	sorted	Hung.	Δ
<i>set-pred head</i>				
resunet	mse	0.926	0.926	0.000
resunet	huber	0.945	0.945	0.000
resunet	Charb	0.991	0.991	0.000
resunet	p-L1	0.992	0.988	−0.004
att-unet	mse	0.807	0.671	−0.136
att-unet	huber	0.929	0.922	−0.007
att-unet	Charb	0.960	0.962	+0.003
att-unet	p-L1	0.942	0.942	0.000
<i>conv head (param losses only)</i>				
resunet	p-MSE	0.937	0.924	−0.013
resunet	p-Hub	0.961	0.961	0.000
resunet	p-L1	0.982	0.982	−0.001
att-unet	p-MSE	0.855	0.849	−0.006
att-unet	p-Hub	0.853	0.849	−0.003
att-unet	p-L1	0.936	0.921	−0.015

composite score; Δ = Hungarian − sorted — zero to slightly negative in all 14 pairs, never a gain. The lone −0.136 (Attention mse) is that loss's instability, not matching.

Winner vs baseline

metric	winner	baseline	Δ
	UNet-skip · L1-curve	UNet · param-MSE	
curve R^2 $\uparrow$	0.756 $\pm .012$	0.580 $\pm .016$	+0.176
PSNR [dB] $\uparrow$	52.8 $\pm .2$	50.4 $\pm .2$	+2.4
SSIM elevation $\uparrow$	0.968 $\pm .000$	0.960 $\pm .000$	+0.007
SSIM range $\uparrow$	0.950 $\pm .002$	0.934 $\pm .004$	+0.016
SSIM azimuth $\uparrow$	0.961 $\pm .001$	0.949 $\pm .001$	+0.012
per-pixel R^2 $\uparrow$	0.658 $\pm .054$	-0.106 $\pm .299$	+0.765
profile cosine $\uparrow$	0.924 $\pm .007$	0.829 $\pm .065$	+0.095
matched F1 $\uparrow$	0.889 $\pm .002$	0.856 $\pm .004$	+0.034
matched recall $\uparrow$	0.886 $\pm .006$	0.894 $\pm .002$	-0.008
matched precision $\uparrow$	0.893 $\pm .005$	0.820 $\pm .006$	+0.073
count exact $\uparrow$	0.861 $\pm .002$	0.845 $\pm .008$	+0.016
matched μ MAE $\downarrow$	1.49 $\pm .04$	2.19 $\pm .17$	-0.70
matched σ MAE $\downarrow$	0.77 $\pm .04$	0.81 $\pm .01$	-0.04
matched a MAE $\downarrow$	0.241 $\pm .011$	0.307 $\pm .011$	-0.066
peak err mean $\downarrow$	1.29 $\pm .05$	1.67 $\pm .16$	-0.38

runs: UNet-skip / UNet · set-pred · Hungarian · $K=2$ · hv flips · presence A · patch 64×32 · five seeds each; held-out validation band. Cells: mean $\pm$ seed std; **bold** = best mean in row by the metric's arrow, marked only where best and worst differ by more than 5%. Δ = winner – baseline on means; **green** = improvement, **red** = regression, coloured only past combined seed noise $\sqrt{s_w^2 + s_b^2}$, grey otherwise. μ , σ , peak in elevation units; a in linear amplitude; cosine and per-pixel R^2 are per-pixel means; count exact = fraction of pixels with the predicted scatterer count correct.

Winner under the microscope: structural fidelity

PENDING — tomogram range slice: GT / prediction / error (clipped at p99), plus SSIM per axis — regenerate from the board winner's inference run 20260720_193330 (UNet-skip · L1-curve, seed 0; five-seed SSIM 0.968 / 0.950 / 0.961 on elevation / range / azimuth)

- Structure is faithfully reconstructed along all three axes; per-slice SSIM in the figure title.

Winner under the microscope: profile reconstruction

PENDING — best-fit and worst-fit profile overlays: GT, prediction, individual Gaussian components, signed residual — from the board winner's inference run 20260720_193330 (UNet-skip · L1-curve, seed 0)

- Best-fit pixels are reconstructed almost exactly.
- Errors concentrate on closely-spaced or low-amplitude components — the heavy negative tail behind the per-pixel mean.

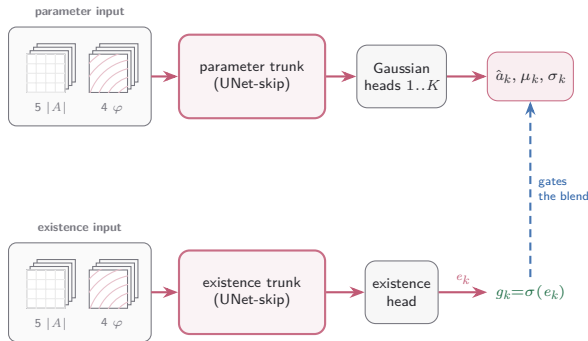
Failure modes, stated before anyone asks

Accuracy is strongest for the dominant low slots and degrades where valid pixels are sparse (June winner run; regenerate from the board winner's run 20260720_193330):

slot	μ MAE	σ MAE	n valid
0	1.3	0.6	12.6 M
1	3.7	0.7	6.6 M
2	12.0	1.3	2.0 M
3	14.7	1.1	1.2 M
4	10.8	1.3	0.7 M

- Inner-slot placeholder recall is the weakest detection link (overall F1 0.834, precision 0.967, recall 0.733).
- Ties back to Act III: the slot imbalance is mitigated, not solved.

The dual model — two trunks, one head



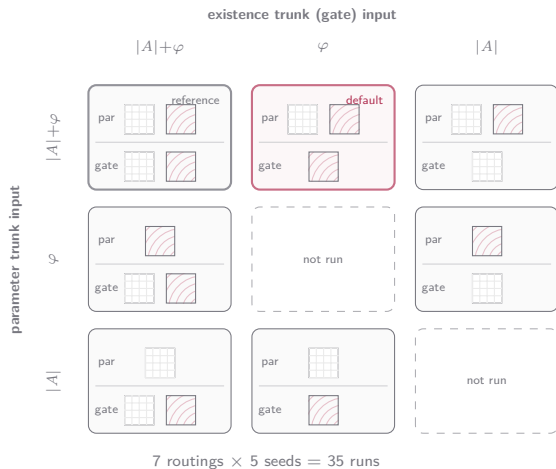
$$\text{each gate: } a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$$

the set-prediction head of Act III, unchanged — only the source of each embedding splits

The gated head asks two questions per slot: *is a scatterer there* (g_k) and *what is it* ($\hat{a}_k, \mu_k, \sigma_k$). So far one backbone embedding had to answer both.

- **Split the encoder, keep the head:** two independent UNet-skip trunks — the Gaussian heads read one embedding, the existence head the other. Gate blend and Hungarian matching stay as they are.
- **Each trunk picks its own stack:** the channel groups $|A|$ (5 amplitudes) and φ (4 interferograms) are selected per trunk — which modality reaches which head becomes an architectural choice.
- **Why route:** the input ablation split the labour — phase fires the slots, the full stack calibrates the parameters. The dual model can hand each head the input its job needs.
- **Default routing:** the full stack $\rightarrow$ parameter trunk, φ alone $\rightarrow$ existence trunk.

Routing the trunks — which stack in front of which head?



The trunks stay identical twins — every arm holds ≈ 31.2 M parameters — so the arms differ only in what each trunk sees.

- **The grid:** parameter-trunk input $\times$ gate input over $\{|A|+\varphi, \varphi, |A|\}$; the two single-modality diagonal cells are omitted.
- **H1 — the parameter trunk needs phase:** with $|A|$ alone in front of the parameters the curve should collapse, and no gate input can buy it back.
- **H2 — the gate runs on phase alone:** the default routing should match the all-channels reference everywhere.
- **H3 — the reversed default pays on the curve:** $\varphi \rightarrow$ parameters should keep detection but give up reconstruction, mirroring the ablation's phase-only arm.

The verdict — the parameter trunk sets the tier, the gate input is nearly free

metric	params $ A +\varphi$			params φ		params $ A $	
	$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
<i>training (validation)</i>							
val loss ↓	0.189 ±.001	0.188 ±.001	0.189 ±.001	0.196 ±.001	0.197 ±.001	0.222 ±.002	0.224 ±.001
<i>reconstruction (curve)</i>							
curve R^2 ↑	0.738 ±.006	0.730 ±.009	0.741 ±.007	0.690 ±.010	0.685 ±.009	0.489 ±.013	0.493 ±.017
curve MAE ↓	0.037 ±.000	0.038 ±.001	0.037 ±.000	0.039 ±.000	0.040 ±.000	0.054 ±.000	0.055 ±.001
PSNR [dB] ↑	52.5 ±.1	52.3 ±.1	52.5 ±.1	51.8 ±.1	51.7 ±.1	49.6 ±.1	49.6 ±.1
SSIM elev ↑	0.967 ±.000	0.967 ±.000	0.967 ±.000	0.966 ±.000	0.965 ±.000	0.956 ±.000	0.955 ±.001
SSIM range ↑	0.951 ±.000	0.951 ±.001	0.951 ±.000	0.949 ±.000	0.948 ±.001	0.927 ±.001	0.926 ±.001
SSIM azim ↑	0.961 ±.000	0.960 ±.001	0.961 ±.000	0.959 ±.000	0.958 ±.000	0.943 ±.000	0.942 ±.001
profile cos med ↑	0.967 ±.002	0.966 ±.007	0.967 ±.002	0.961 ±.007	0.963 ±.004	0.887 ±.009	0.881 ±.007
<i>detection (matched)</i>							
precision ↑	0.905 ±.001	0.907 ±.002	0.901 ±.003	0.892 ±.006	0.892 ±.011	0.836 ±.015	0.843 ±.014
recall ↑	0.892 ±.002	0.884 ±.004	0.890 ±.004	0.893 ±.005	0.886 ±.004	0.862 ±.004	0.852 ±.003
matched F1 ↑	0.898 ±.001	0.895 ±.003	0.896 ±.003	0.893 ±.003	0.889 ±.004	0.849 ±.008	0.848 ±.006
count exact ↑	0.914 ±.001	0.907 ±.003	0.909 ±.006	0.908 ±.003	0.904 ±.002	0.870 ±.023	0.864 ±.014
peak err med ↓	0.27 ±.37	0.13 ±.30	0.40 ±.37	0.40 ±.37	0.54 ±.30	0.67 ±.00	0.67 ±.00
peak err p95 ↓	5.4 ±.0	6.0 ±.5	5.5 ±.3	5.0 ±.4	5.8 ±.6	7.5 ±.6	8.1 ±.0

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric twin trunks, ≈ 31.2 M params per arm. Column group = parameter-trunk input, sub-column = existence (gate) trunk input; $|A|+\varphi$ = 5 amplitudes + 4 interferograms. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px. **bold** = best in row, marked only where best and worst differ by more than 5%; **red** = the curve price of an amplitude-only parameter trunk; matched = Hungarian pairing. Peak err med/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

Dual routing by scatterer count — detection survives without phase, position does not

metric	params $ A +\varphi$			params φ		params $ A $	
	$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
precision $\uparrow$	0.905 $\pm$.001	0.907 $\pm$.002	0.901 $\pm$.003	0.892 $\pm$.006	0.892 $\pm$.011	0.836 $\pm$.015	0.843 $\pm$.014
$k=1$	0.949 $\pm$.002	0.952 $\pm$.002	0.944 $\pm$.005	0.938 $\pm$.007	0.940 $\pm$.009	0.889 $\pm$.025	0.898 $\pm$.021
$k=2$	0.831 $\pm$.001	0.831 $\pm$.004	0.829 $\pm$.002	0.816 $\pm$.005	0.812 $\pm$.014	0.745 $\pm$.006	0.749 $\pm$.006
recall $\uparrow$	0.892 $\pm$.002	0.884 $\pm$.004	0.890 $\pm$.004	0.893 $\pm$.005	0.886 $\pm$.004	0.862 $\pm$.004	0.852 $\pm$.003
$k=1$	0.997 $\pm$.001	0.992 $\pm$.002	0.996 $\pm$.002	0.998 $\pm$.000	0.998 $\pm$.001	0.991 $\pm$.002	0.980 $\pm$.007
$k=2$	0.743 $\pm$.004	0.729 $\pm$.006	0.740 $\pm$.008	0.743 $\pm$.011	0.727 $\pm$.010	0.680 $\pm$.007	0.670 $\pm$.008
<i>count acc pred k</i>							
pred $k=1 \uparrow$	0.953 $\pm$.003	0.947 $\pm$.003	0.953 $\pm$.002	0.941 $\pm$.008	0.939 $\pm$.005	0.958 $\pm$.009	0.946 $\pm$.002
pred $k=2 \uparrow$	0.846 $\pm$.003	0.851 $\pm$.005	0.834 $\pm$.012	0.820 $\pm$.015	0.822 $\pm$.017	0.718 $\pm$.051	0.732 $\pm$.045
μ MAE $\downarrow$	1.63 $\pm$.01	1.59 $\pm$.04	1.63 $\pm$.02	1.73 $\pm$.06	1.74 $\pm$.16	2.35 $\pm$.07	2.31 $\pm$.03
$k=1$	0.33 $\pm$.01	0.33 $\pm$.03	0.33 $\pm$.02	0.37 $\pm$.03	0.36 $\pm$.03	0.79 $\pm$.06	0.79 $\pm$.03
$k=2$	3.70 $\pm$.02	3.62 $\pm$.06	3.70 $\pm$.04	3.85 $\pm$.08	3.93 $\pm$.31	4.77 $\pm$.16	4.68 $\pm$.05
σ MAE $\downarrow$	0.694 $\pm$.005	0.692 $\pm$.006	0.691 $\pm$.007	0.783 $\pm$.032	0.763 $\pm$.050	0.828 $\pm$.037	0.851 $\pm$.007
$k=1$	0.185 $\pm$.002	0.189 $\pm$.006	0.185 $\pm$.003	0.221 $\pm$.013	0.210 $\pm$.013	0.235 $\pm$.013	0.246 $\pm$.009
$k=2$	1.503 $\pm$.009	1.503 $\pm$.014	1.495 $\pm$.018	1.660 $\pm$.056	1.640 $\pm$.090	1.746 $\pm$.072	1.798 $\pm$.011
a MAE $\downarrow$	0.181 $\pm$.003	0.182 $\pm$.002	0.184 $\pm$.002	0.185 $\pm$.004	0.191 $\pm$.007	0.192 $\pm$.015	0.191 $\pm$.004
$k=1$	0.056 $\pm$.001	0.056 $\pm$.002	0.059 $\pm$.001	0.061 $\pm$.003	0.063 $\pm$.003	0.066 $\pm$.009	0.066 $\pm$.004
$k=2$	0.380 $\pm$.006	0.385 $\pm$.005	0.382 $\pm$.003	0.380 $\pm$.009	0.395 $\pm$.019	0.387 $\pm$.023	0.387 $\pm$.006

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric twin trunks. Column group = parameter-trunk input, sub-column = gate-trunk input. All precision / recall / μ / σ / a rows are matched (Hungarian pairing). Cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel. **bold** = best in row, marked only where best and worst differ by more than 5%; **green** = the sharpest localisation, both on a full-parameter trunk, **red** = position dying without phase. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

Dual routing, predicted statistics vs the GT — calibration follows the parameter trunk

mean	GT	params $ A +\varphi$			params φ		params $ A $	
		$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
slot 0 — strongest scatterer; GT-active on every pixel								
active frac	1.000	0.973 $\pm$.002	0.968 $\pm$.002	0.973 $\pm$.003	0.983 $\pm$.005	0.977 $\pm$.006	0.978 $\pm$.007	0.969 $\pm$.006
a — active	0.77	0.72 $\pm$.01	0.76 $\pm$.01	0.72 $\pm$.01	0.72 $\pm$.02	0.72 $\pm$.01	0.73 $\pm$.01	0.73 $\pm$.01
μ — active	-4.89	-4.57 $\pm$.04	-4.53 $\pm$.04	-4.54 $\pm$.05	-4.56 $\pm$.04	-4.52 $\pm$.07	-4.46 $\pm$.10	-4.49 $\pm$.08
σ — active	1.76	1.35 $\pm$.02	1.34 $\pm$.01	1.37 $\pm$.02	1.39 $\pm$.03	1.36 $\pm$.03	1.38 $\pm$.03	1.37 $\pm$.03
slot 1 — second scatterer; GT-active on 26% of pixels								
active frac	0.260	0.269 $\pm$.003	0.260 $\pm$.002	0.272 $\pm$.004	0.278 $\pm$.009	0.274 $\pm$.015	0.322 $\pm$.025	0.305 $\pm$.021
a — active	1.43	1.43 $\pm$.01	1.47 $\pm$.03	1.42 $\pm$.03	1.44 $\pm$.06	1.38 $\pm$.05	1.39 $\pm$.07	1.44 $\pm$.02
a — inact.	0.00	0.87 $\pm$.04	0.90 $\pm$.06	0.82 $\pm$.09	0.78 $\pm$.05	0.75 $\pm$.15	0.53 $\pm$.10	0.57 $\pm$.10
μ — active	11.34	7.66 $\pm$.13	7.70 $\pm$.24	7.72 $\pm$.26	7.68 $\pm$.37	8.15 $\pm$.38	6.96 $\pm$.31	7.04 $\pm$.26
μ — inact.	—	0.82 $\pm$.16	0.94 $\pm$.38	0.50 $\pm$.48	0.67 $\pm$.57	0.44 $\pm$.93	-0.35 $\pm$ 1.18	-0.86 $\pm$.87
σ — active	3.53	3.23 $\pm$.06	3.21 $\pm$.06	3.24 $\pm$.02	3.29 $\pm$.10	3.28 $\pm$.06	3.14 $\pm$.06	3.10 $\pm$.05
σ — inact.	—	1.81 $\pm$.02	1.79 $\pm$.04	1.79 $\pm$.09	1.89 $\pm$.02	1.83 $\pm$.07	2.00 $\pm$.26	1.85 $\pm$.26
distribution width — std over GT-active pixels								
a — slot 0	1.48	1.28 $\pm$.03	1.35 $\pm$.02	1.25 $\pm$.03	1.20 $\pm$.04	1.18 $\pm$.03	1.30 $\pm$.03	1.33 $\pm$.04
μ — slot 0	3.30	2.33 $\pm$.07	2.22 $\pm$.09	2.35 $\pm$.07	2.32 $\pm$.09	2.46 $\pm$.21	2.38 $\pm$.29	2.35 $\pm$.41
σ — slot 0	2.84	1.15 $\pm$.04	1.15 $\pm$.04	1.16 $\pm$.04	1.23 $\pm$.09	1.15 $\pm$.06	1.12 $\pm$.08	1.14 $\pm$.14
a — slot 1	1.45	1.28 $\pm$.02	1.29 $\pm$.02	1.26 $\pm$.05	1.26 $\pm$.07	1.20 $\pm$.03	1.24 $\pm$.05	1.28 $\pm$.01
μ — slot 1	16.24	8.85 $\pm$.09	8.62 $\pm$.37	8.78 $\pm$.34	8.63 $\pm$.38	9.56 $\pm$.75	7.81 $\pm$.28	7.71 $\pm$.14
σ — slot 1	4.01	1.72 $\pm$.03	1.69 $\pm$.04	1.67 $\pm$.05	1.58 $\pm$.08	1.59 $\pm$.04	1.32 $\pm$.04	1.26 $\pm$.05

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric twin trunks. Column group = parameter-trunk input, sub-column = gate-trunk input. Five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = calibration following the parameter trunk; inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; a in linear amplitude.

Dual routing at $K=5$ — the parameter trunk still sets the tier, the gate turns conservative

metric	params $ A +\varphi$			params φ		params $ A $	
	$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
<i>training (validation)</i>							
val loss ↓	0.130 ±.000	0.129 ±.000	0.129 ±.000	0.133 ±.000	0.135 ±.001	0.143 ±.001	0.148 ±.002
<i>reconstruction (curve)</i>							
curve R^2 ↑	0.679 ±.010	0.676 ±.005	0.680 ±.014	0.624 ±.007	0.635 ±.015	0.458 ±.011	0.475 ±.014
curve MAE ↓	0.046 ±.000	0.046 ±.001	0.045 ±.000	0.047 ±.000	0.047 ±.001	0.062 ±.001	0.062 ±.002
PSNR [dB] ↑	51.0 ±.1	51.0 ±.1	51.0 ±.2	50.3 ±.1	50.5 ±.2	48.7 ±.1	48.9 ±.1
SSIM elev ↑	0.961 ±.000	0.961 ±.000	0.962 ±.000	0.961 ±.000	0.960 ±.001	0.953 ±.000	0.953 ±.001
SSIM range ↑	0.940 ±.001	0.940 ±.001	0.940 ±.001	0.939 ±.001	0.938 ±.003	0.921 ±.003	0.920 ±.004
SSIM azim ↑	0.954 ±.000	0.954 ±.000	0.955 ±.000	0.953 ±.000	0.953 ±.001	0.939 ±.001	0.940 ±.002
profile cos med ↑	0.922 ±.011	0.932 ±.005	0.927 ±.005	0.920 ±.008	0.922 ±.015	0.795 ±.041	0.791 ±.065
<i>detection (matched)</i>							
precision ↑	0.917 ±.006	0.917 ±.003	0.915 ±.004	0.916 ±.010	0.900 ±.011	0.891 ±.012	0.875 ±.026
recall ↑	0.666 ±.008	0.671 ±.018	0.670 ±.026	0.697 ±.025	0.673 ±.066	0.661 ±.081	0.706 ±.050
matched F1 ↑	0.772 ±.007	0.774 ±.011	0.774 ±.017	0.791 ±.016	0.769 ±.046	0.757 ±.056	0.780 ±.026
count exact ↑	0.632 ±.012	0.638 ±.022	0.635 ±.035	0.669 ±.030	0.628 ±.085	0.648 ±.103	0.675 ±.048
under ↓	0.329 ±.011	0.323 ±.023	0.325 ±.037	0.289 ±.031	0.323 ±.084	0.310 ±.109	0.251 ±.070
over ↓	0.039 ±.003	0.039 ±.003	0.040 ±.002	0.041 ±.008	0.049 ±.004	0.043 ±.011	0.074 ±.040
peak err med ↓	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	1.07 ±.37	1.07 ±.37
peak err p95 ↓	14.1 ±.0	14.1 ±.0	14.1 ±.0	13.7 ±.6	13.2 ±.2.1	13.2 ±.1.5	13.6 ±.1.1

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric parity-twin trunks, ≈ 31.2 M params per arm. Column group = parameter-trunk input, sub-column = existence (gate) trunk input; $|A|+\varphi$ = 5 amplitudes + 4 interferograms. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px, $K_{\max}=5$ labels. **bold** = best in row, marked only where best and worst differ by more than 5%; **red** = the curve price of an amplitude-only parameter trunk. under / over = predicted count below / above the GT count: the dual misses tail scatterers (under 0.25–0.33, over ≤ 0.07) — the opposite failure of the $K=5$ single-trunk arms. Peak err med/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

The $K=5$ dual by scatterer count — $k=1$ no longer saturates, recall fades up the ladder

metric	params $ A +\varphi$			params φ		params $ A $	
	$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
precision $\uparrow$	0.917 $\pm$.006	0.917 $\pm$.003	0.915 $\pm$.004	0.916 $\pm$.010	0.900 $\pm$.011	0.891 $\pm$.012	0.875 $\pm$.026
$k=1$	0.936 $\pm$.005	0.935 $\pm$.004	0.935 $\pm$.004	0.934 $\pm$.013	0.908 $\pm$.015	0.921 $\pm$.013	0.890 $\pm$.038
$k=2$	0.912 $\pm$.008	0.911 $\pm$.004	0.913 $\pm$.009	0.910 $\pm$.004	0.910 $\pm$.007	0.848 $\pm$.007	0.856 $\pm$.009
$k=3$	0.869 $\pm$.010	0.871 $\pm$.010	0.864 $\pm$.016	0.869 $\pm$.006	0.872 $\pm$.013	0.842 $\pm$.005	0.848 $\pm$.009
$k=4$	0.819 $\pm$.013	0.820 $\pm$.019	0.805 $\pm$.028	0.815 $\pm$.014	0.820 $\pm$.018	0.809 $\pm$.009	0.812 $\pm$.013
$k=5$	0.794 $\pm$.015	0.791 $\pm$.028	0.769 $\pm$.042	0.780 $\pm$.022	0.783 $\pm$.020	0.787 $\pm$.017	0.790 $\pm$.016
recall $\uparrow$	0.666 $\pm$.008	0.671 $\pm$.018	0.670 $\pm$.026	0.697 $\pm$.025	0.673 $\pm$.066	0.661 $\pm$.081	0.706 $\pm$.050
$k=1$	0.757 $\pm$.014	0.766 $\pm$.031	0.761 $\pm$.047	0.807 $\pm$.038	0.768 $\pm$.109	0.787 $\pm$.143	0.858 $\pm$.093
$k=2$	0.701 $\pm$.008	0.700 $\pm$.007	0.708 $\pm$.006	0.707 $\pm$.025	0.704 $\pm$.023	0.632 $\pm$.011	0.657 $\pm$.011
$k=3$	0.508 $\pm$.005	0.507 $\pm$.006	0.509 $\pm$.004	0.510 $\pm$.010	0.506 $\pm$.008	0.472 $\pm$.003	0.477 $\pm$.009
$k=4$	0.358 $\pm$.008	0.356 $\pm$.005	0.360 $\pm$.004	0.362 $\pm$.009	0.359 $\pm$.010	0.331 $\pm$.006	0.332 $\pm$.007
$k=5$	0.239 $\pm$.008	0.239 $\pm$.005	0.242 $\pm$.006	0.257 $\pm$.011	0.251 $\pm$.016	0.219 $\pm$.008	0.219 $\pm$.008
count acc pred $k \uparrow$							
pred $k=1$	0.885 $\pm$.005	0.887 $\pm$.005	0.887 $\pm$.006	0.888 $\pm$.012	0.868 $\pm$.020	0.873 $\pm$.018	0.884 $\pm$.006
pred $k=2$	0.496 $\pm$.005	0.488 $\pm$.006	0.493 $\pm$.003	0.485 $\pm$.007	0.469 $\pm$.011	0.454 $\pm$.023	0.413 $\pm$.063
pred $k=3$	0.252 $\pm$.012	0.236 $\pm$.015	0.243 $\pm$.008	0.239 $\pm$.009	0.164 $\pm$.054	0.207 $\pm$.024	0.147 $\pm$.067
pred $k=4$	0.198 $\pm$.014	0.188 $\pm$.016	0.210 $\pm$.013	0.198 $\pm$.020	0.110 $\pm$.066	0.180 $\pm$.052	0.169 $\pm$.027
pred $k=5$	0.198 $\pm$.039	0.196 $\pm$.071	0.222 $\pm$.024	0.183 $\pm$.045	0.142 $\pm$.072	0.129 $\pm$.067	0.306 $\pm$.393

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric parity-twin trunks. Column group = parameter-trunk input, sub-column = gate-trunk input. All precision / recall rows are matched (Hungarian pairing). Cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel. **bold** = best in row, marked only where best and worst differ by more than 5%; **red** = the two-scatterer price of a phase-free parameter trunk, and the $k=1$ seed lottery (coloured $\pm$ spreads). Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k .

The $K=5$ dual components — position pays for the crowd, and doubly without phase

metric	params $ A +\varphi$			params φ		params $ A $	
	$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
μ MAE $\downarrow$	1.16 $\pm$.05	1.15 $\pm$.03	1.17 $\pm$.03	1.19 $\pm$.05	1.18 $\pm$.08	1.73 $\pm$.12	1.74 $\pm$.18
$k=1$	0.48 $\pm$.03	0.45 $\pm$.01	0.46 $\pm$.02	0.51 $\pm$.02	0.49 $\pm$.02	1.02 $\pm$.09	1.13 $\pm$.23
$k=2$	1.81 $\pm$.03	1.88 $\pm$.05	1.83 $\pm$.06	1.90 $\pm$.03	1.91 $\pm$.08	2.76 $\pm$.06	2.69 $\pm$.06
$k=3$	2.55 $\pm$.18	2.48 $\pm$.14	2.51 $\pm$.09	2.54 $\pm$.12	2.43 $\pm$.13	2.92 $\pm$.04	2.90 $\pm$.15
$k=4$	3.26 $\pm$.26	3.20 $\pm$.27	3.34 $\pm$.32	3.26 $\pm$.24	3.05 $\pm$.21	3.30 $\pm$.10	3.30 $\pm$.25
$k=5$	3.52 $\pm$.23	3.48 $\pm$.45	3.83 $\pm$.54	3.55 $\pm$.31	3.49 $\pm$.36	3.45 $\pm$.20	3.43 $\pm$.26
σ MAE $\downarrow$	0.60 $\pm$.01	0.60 $\pm$.01	0.60 $\pm$.02	0.63 $\pm$.02	0.64 $\pm$.03	0.70 $\pm$.07	0.67 $\pm$.04
$k=1$	0.27 $\pm$.01	0.27 $\pm$.00	0.27 $\pm$.01	0.28 $\pm$.01	0.29 $\pm$.02	0.33 $\pm$.03	0.32 $\pm$.03
$k=2$	0.88 $\pm$.01	0.89 $\pm$.01	0.89 $\pm$.01	0.94 $\pm$.01	0.95 $\pm$.02	1.14 $\pm$.02	1.11 $\pm$.02
$k=3$	1.31 $\pm$.02	1.33 $\pm$.01	1.33 $\pm$.02	1.37 $\pm$.02	1.36 $\pm$.02	1.50 $\pm$.02	1.47 $\pm$.02
$k=4$	1.65 $\pm$.02	1.66 $\pm$.02	1.68 $\pm$.03	1.76 $\pm$.03	1.75 $\pm$.04	1.75 $\pm$.02	1.74 $\pm$.04
$k=5$	1.83 $\pm$.02	1.84 $\pm$.03	1.85 $\pm$.03	2.02 $\pm$.05	2.02 $\pm$.09	1.90 $\pm$.04	1.91 $\pm$.10
a MAE $\downarrow$	0.212 $\pm$.005	0.198 $\pm$.002	0.205 $\pm$.011	0.199 $\pm$.007	0.225 $\pm$.023	0.198 $\pm$.014	0.201 $\pm$.017
$k=1$	0.076 $\pm$.004	0.073 $\pm$.003	0.077 $\pm$.008	0.079 $\pm$.005	0.094 $\pm$.012	0.077 $\pm$.009	0.076 $\pm$.011
$k=2$	0.387 $\pm$.013	0.366 $\pm$.013	0.371 $\pm$.015	0.370 $\pm$.008	0.400 $\pm$.017	0.375 $\pm$.017	0.399 $\pm$.032
$k=3$	0.464 $\pm$.013	0.431 $\pm$.008	0.440 $\pm$.012	0.436 $\pm$.007	0.465 $\pm$.027	0.433 $\pm$.014	0.466 $\pm$.029
$k=4$	0.498 $\pm$.016	0.448 $\pm$.008	0.466 $\pm$.015	0.440 $\pm$.013	0.477 $\pm$.039	0.442 $\pm$.018	0.475 $\pm$.029
$k=5$	0.498 $\pm$.023	0.440 $\pm$.018	0.456 $\pm$.017	0.403 $\pm$.017	0.455 $\pm$.059	0.414 $\pm$.023	0.446 $\pm$.035

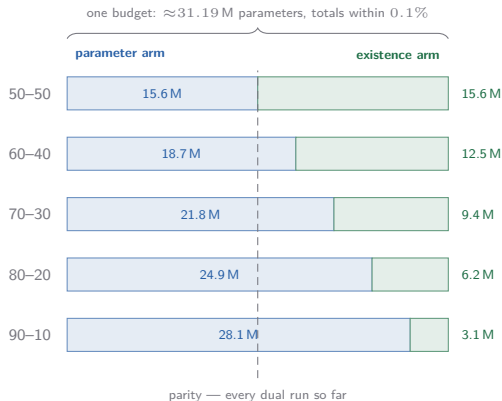
runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric parity-twin trunks. Column group = parameter-trunk input, sub-column = gate-trunk input. Cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only — low-recall arms are graded on an easier subset. **bold** = best in row, marked only where best and worst differ by more than 5%; **red** = position paying double without phase in the parameter trunk. μ , σ in elevation units; a in linear amplitude.

The $K=5$ dual, predicted statistics vs the GT — the always-active slot fires on three quarters of the pixels

		params $ A +\varphi$			params φ		params $ A $	
mean	GT	$ A +\varphi$	φ	$ A $	$ A +\varphi$	$ A $	$ A +\varphi$	φ
<i>predicted active fraction — share of pixels where the slot fires</i>								
slot 0	1.000	0.762 $\pm$.011	0.768 $\pm$.020	0.763 $\pm$.036	0.802 $\pm$.029	0.769 $\pm$.086	0.790 $\pm$.107	0.847 $\pm$.068
slot 1	0.239	0.160 $\pm$.005	0.161 $\pm$.003	0.169 $\pm$.016	0.169 $\pm$.013	0.169 $\pm$.002	0.168 $\pm$.014	0.165 $\pm$.010
slot 2	0.085	0.037 $\pm$.006	0.046 $\pm$.007	0.044 $\pm$.011	0.046 $\pm$.017	0.038 $\pm$.006	0.038 $\pm$.012	0.036 $\pm$.015
slot 3	0.035	0.009 $\pm$.002	0.008 $\pm$.002	0.016 $\pm$.010	0.013 $\pm$.009	0.036 $\pm$.017	0.013 $\pm$.012	0.023 $\pm$.017
slot 4	0.015	0.030 $\pm$.009	0.022 $\pm$.013	0.014 $\pm$.010	0.016 $\pm$.019	0.016 $\pm$.013	0.008 $\pm$.008	0.040 $\pm$.050
<i>a — predicted mean over GT-active pixels</i>								
slot 0	0.84	0.94 $\pm$.02	0.98 $\pm$.04	0.95 $\pm$.04	0.90 $\pm$.04	0.91 $\pm$.09	0.93 $\pm$.13	0.86 $\pm$.06
slot 1	1.67	1.82 $\pm$.04	1.84 $\pm$.04	1.82 $\pm$.06	1.81 $\pm$.08	1.76 $\pm$.11	1.78 $\pm$.04	1.75 $\pm$.03
slot 2	1.41	1.81 $\pm$.03	1.86 $\pm$.05	1.78 $\pm$.11	1.81 $\pm$.09	1.80 $\pm$.09	1.91 $\pm$.06	1.86 $\pm$.09
slot 3	0.97	1.29 $\pm$.06	1.29 $\pm$.14	1.15 $\pm$.17	1.13 $\pm$.05	1.32 $\pm$.07	1.75 $\pm$.24	1.58 $\pm$.11
slot 4	0.66	0.65 $\pm$.10	0.51 $\pm$.07	0.55 $\pm$.06	0.24 $\pm$.03	0.57 $\pm$.09	1.09 $\pm$.57	0.61 $\pm$.31
<i>a — predicted mean over GT-inactive pixels (leakage)</i>								
slot 1	0.00	0.79 $\pm$.86	0.50 $\pm$.86	0.92 $\pm$.67	0.70 $\pm$.85	0.59 $\pm$.67	1.27 $\pm$.57	0.59 $\pm$.75
slot 2	0.00	0.69 $\pm$.60	1.31 $\pm$.71	0.98 $\pm$.72	0.75 $\pm$.86	0.50 $\pm$.67	0.99 $\pm$.83	0.09 $\pm$.03
slot 3	0.00	0.33 $\pm$.19	0.26 $\pm$.07	0.82 $\pm$.64	0.49 $\pm$.82	0.92 $\pm$.53	0.42 $\pm$.77	0.84 $\pm$.68
slot 4	0.00	1.53 $\pm$.16	1.26 $\pm$.63	0.91 $\pm$.84	0.62 $\pm$.76	0.68 $\pm$.73	0.41 $\pm$.70	0.76 $\pm$.83
<i>distribution width — std over GT-active pixels</i>								
a — slot 0	1.59	1.48 $\pm$.01	1.60 $\pm$.04	1.53 $\pm$.06	1.41 $\pm$.04	1.38 $\pm$.05	1.50 $\pm$.11	1.45 $\pm$.03
μ — slot 0	3.02	2.41 $\pm$.06	2.44 $\pm$.08	2.51 $\pm$.10	2.41 $\pm$.09	2.51 $\pm$.05	2.35 $\pm$.08	2.22 $\pm$.16
σ — slot 0	1.72	1.03 $\pm$.04	1.03 $\pm$.06	1.04 $\pm$.06	1.04 $\pm$.06	1.06 $\pm$.09	0.97 $\pm$.08	0.91 $\pm$.06
a — slot 1	1.60	1.38 $\pm$.03	1.43 $\pm$.04	1.41 $\pm$.03	1.34 $\pm$.05	1.37 $\pm$.05	1.37 $\pm$.03	1.34 $\pm$.06
μ — slot 1	11.09	8.96 $\pm$.40	8.73 $\pm$.21	8.77 $\pm$.22	8.78 $\pm$.16	8.74 $\pm$.29	8.21 $\pm$.05	7.75 $\pm$.42
σ — slot 1	2.30	1.40 $\pm$.02	1.40 $\pm$.04	1.43 $\pm$.03	1.32 $\pm$.03	1.33 $\pm$.03	1.31 $\pm$.07	1.29 $\pm$.03

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · symmetric parity-twin trunks. Column group = parameter-trunk input, sub-column = gate-trunk input. Five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%. The always-active slot 0 fires on 0.76–0.85 of pixels and slots 2–3 sit below their GT activity — the $K=5$ dual gate under-counts rather than over-fires. Width rows: std of the prediction over GT-active pixels vs the GT's own std, shown for slots 0–1 (tail slots in the report). μ , σ in elevation units; a in linear amplitude.

Splitting the budget — how much of the model does detection need?



5 splits $\times$ 5 seeds, at $K=2$ and $K=5 = 50$ runs; width scales like $\sqrt{\text{share}}$:
the 10% ladder (24-48-84-156) is half the parity width

The routing block fixed *what* each trunk sees. This experiment fixes the inputs and asks how one budget should be divided between the two arms.

- **One budget, five splits:** the parity total is redistributed at fixed routing (the full stack in front of both trunks); ladders keep four levels, only width moves, and every realized split sits within 0.01 points of its label.
- **An arm is trunk + its heads:** every parameter belongs to exactly one arm; the 50-50 anchor is the routing block's all-channels reference arm.
- **H1 — detection is cheap:** count and calibration hold to 80/20 or 90/10 and the freed capacity buys μ/σ — a better operating point than parity.
- **H2 — parity is load-bearing:** count collapses as the existence arm shrinks, and the twins stand.
- **H3 — the telling order:** μ/σ degrade before count does — the gate was borrowing parameter-trunk capacity through the shared loss.

The ratio verdict — detection survives on a tenth of the budget

metric	parameter : existence split					
	50–50	60–40	70–30	80–20	90–10	single
<i>training (validation)</i>						
val loss ↓	0.189 ±.001	0.189 ±.001	0.189 ±.000	0.187 ±.000	0.188 ±.001	0.188 ±.000
<i>reconstruction (curve)</i>						
curve R^2 ↑	0.738 ±.006	0.735 ±.009	0.739 ±.012	0.734 ±.018	0.748 ±.010	0.741 ±.016
curve MAE ↓	0.0373 ±.0004	0.0376 ±.0010	0.0375 ±.0005	0.0378 ±.0014	0.0370 ±.0002	0.0372 ±.0006
PSNR [dB] ↑	52.5 ±.1	52.4 ±.1	52.5 ±.2	52.4 ±.3	52.6 ±.2	52.5 ±.3
SSIM elev ↑	0.9668 ±.0002	0.9667 ±.0006	0.9667 ±.0003	0.9665 ±.0010	0.9669 ±.0001	0.9669 ±.0004
SSIM range ↑	0.9514 ±.0005	0.9510 ±.0013	0.9508 ±.0005	0.9508 ±.0015	0.9514 ±.0004	0.9513 ±.0006
SSIM azim ↑	0.9608 ±.0003	0.9605 ±.0010	0.9605 ±.0004	0.9603 ±.0014	0.9609 ±.0002	0.9608 ±.0005
profile cos med ↑	0.967 ±.002	0.966 ±.005	0.966 ±.004	0.969 ±.002	0.969 ±.001	0.967 ±.002
<i>detection (matched)</i>						
precision ↑	0.905 ±.001	0.902 ±.003	0.903 ±.003	0.905 ±.003	0.905 ±.003	0.900 ±.001
recall ↑	0.892 ±.002	0.887 ±.004	0.891 ±.001	0.890 ±.004	0.891 ±.003	0.893 ±.004
matched F1 ↑	0.898 ±.001	0.895 ±.004	0.897 ±.001	0.897 ±.003	0.898 ±.002	0.897 ±.002
count exact ↑	0.914 ±.001	0.909 ±.005	0.912 ±.002	0.910 ±.003	0.910 ±.004	0.917 ±.004
under ↓	0.047 ±.002	0.050 ±.003	0.047 ±.002	0.050 ±.002	0.049 ±.002	0.044 ±.004
over ↓	0.039 ±.001	0.041 ±.003	0.041 ±.003	0.040 ±.003	0.040 ±.004	0.039 ±.002
peak err med ↓	0.27 ±.37	0.27 ±.37	0.40 ±.37	0.13 ±.30	0.00 ±.00	0.13 ±.30
peak err p95 ↓	5.4 ±.0	5.8 ±.6	5.4 ±.0	5.6 ±.4	5.4 ±.0	5.4 ±.0

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks). Column = parameter : existence split of the ≈ 31.19 M parity budget (four-level ladders, only width moves); the 50–50 arm is the routing block's all-channels reference arm. single = the benchmark board's UNet-skip · set-pred · Hungarian · param-L1 arm: one size-matched trunk, 31.35 M params, same seeds and patch. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px. **bold** = best in row, marked only where best and worst differ by more than 5%. 90–10 takes the displayed best on every curve row (each edge inside combined seed noise), the detection rows never leave the 50–50 level, and the single trunk sits on the same tier throughout — its count-exact edge (0.917 vs 0.914) is inside seed noise. Peak err med/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

Ratio by scatterer count — the freed capacity lands in localisation

metric	parameter : existence split					single
	50–50	60–40	70–30	80–20	90–10	
precision $\uparrow$	0.905 $\pm$.001	0.902 $\pm$.003	0.903 $\pm$.003	0.905 $\pm$.003	0.905 $\pm$.003	0.900 $\pm$.001
$k=1$	0.949 $\pm$.002	0.947 $\pm$.004	0.947 $\pm$.004	0.947 $\pm$.004	0.947 $\pm$.005	0.948 $\pm$.002
$k=2$	0.831 $\pm$.001	0.828 $\pm$.004	0.830 $\pm$.002	0.834 $\pm$.004	0.834 $\pm$.002	0.821 $\pm$.002
recall $\uparrow$	0.892 $\pm$.002	0.887 $\pm$.004	0.891 $\pm$.001	0.890 $\pm$.004	0.891 $\pm$.003	0.893 $\pm$.004
$k=1$	0.997 $\pm$.001	0.994 $\pm$.003	0.996 $\pm$.001	0.995 $\pm$.001	0.995 $\pm$.002	0.997 $\pm$.001
$k=2$	0.743 $\pm$.004	0.735 $\pm$.007	0.742 $\pm$.004	0.740 $\pm$.008	0.742 $\pm$.006	0.745 $\pm$.008
count acc pred k						
pred $k=1$ $\uparrow$	0.953 $\pm$.003	0.954 $\pm$.004	0.953 $\pm$.003	0.953 $\pm$.003	0.952 $\pm$.003	0.950 $\pm$.003
pred $k=2$ $\uparrow$	0.846 $\pm$.003	0.839 $\pm$.010	0.841 $\pm$.010	0.841 $\pm$.011	0.841 $\pm$.013	0.847 $\pm$.005
μ MAE $\downarrow$	1.63 $\pm$.01	1.63 $\pm$.02	1.63 $\pm$.01	1.59 $\pm$.02	1.59 $\pm$.02	1.72 $\pm$.04
$k=1$	0.33 $\pm$.01	0.33 $\pm$.02	0.34 $\pm$.02	0.32 $\pm$.01	0.31 $\pm$.01	0.34 $\pm$.01
$k=2$	3.70 $\pm$.02	3.69 $\pm$.03	3.67 $\pm$.03	3.62 $\pm$.03	3.63 $\pm$.05	3.87 $\pm$.07
σ MAE $\downarrow$	0.694 $\pm$.005	0.686 $\pm$.004	0.698 $\pm$.003	0.689 $\pm$.007	0.690 $\pm$.008	0.735 $\pm$.014
$k=1$	0.185 $\pm$.002	0.181 $\pm$.005	0.186 $\pm$.004	0.182 $\pm$.005	0.185 $\pm$.005	0.198 $\pm$.004
$k=2$	1.503 $\pm$.009	1.490 $\pm$.012	1.509 $\pm$.011	1.498 $\pm$.014	1.494 $\pm$.021	1.575 $\pm$.029
a MAE $\downarrow$	0.181 $\pm$.003	0.184 $\pm$.003	0.182 $\pm$.002	0.183 $\pm$.005	0.177 $\pm$.002	0.175 $\pm$.002
$k=1$	0.056 $\pm$.001	0.056 $\pm$.002	0.056 $\pm$.002	0.058 $\pm$.003	0.056 $\pm$.003	0.055 $\pm$.001
$k=2$	0.380 $\pm$.006	0.388 $\pm$.008	0.381 $\pm$.002	0.383 $\pm$.010	0.371 $\pm$.002	0.363 $\pm$.007

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks). Column = parameter:existence split of the parity budget. All rows matched (Hungarian pairing); cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel. single = the benchmark UNet-skip · set-pred · param-L1 single trunk (31.35 M). **bold** = best in row, marked only where best and worst differ by more than 5%; **green** = localisation collecting the freed capacity — the μ gains clear combined seed noise, the amplitude gains just clear it; **red** = the single trunk's localisation price (μ 1.72 vs 1.59–1.63, σ 0.735 vs 0.686–0.698, both past combined noise) while it answers back on amplitude and recall — splitting the encoder buys localisation, at any split. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

Ratio, predicted statistics vs the GT — calibration is split-blind

		parameter : existence split					
mean	GT	50–50	60–40	70–30	80–20	90–10	single
slot 0 — strongest scatterer; GT-active on every pixel							
active frac	1.000	0.973 ±.002	0.970 ±.003	0.974 ±.002	0.971 ±.002	0.972 ±.003	0.976 ±.003
a — active	0.77	0.72 ±.01	0.73 ±.03	0.73 ±.01	0.73 ±.00	0.74 ±.01	0.74 ±.00
μ — active	−4.89	−4.57 ±.04	−4.56 ±.04	−4.57 ±.06	−4.62 ±.04	−4.58 ±.05	−4.52 ±.04
σ — active	1.76	1.35 ±.02	1.38 ±.02	1.38 ±.02	1.36 ±.01	1.38 ±.03	1.38 ±.01
slot 1 — second scatterer; GT-active on 26% of pixels							
active frac	0.260	0.269 ±.003	0.268 ±.003	0.269 ±.005	0.268 ±.005	0.268 ±.004	0.274 ±.003
a — active	1.43	1.43 ±.01	1.43 ±.02	1.43 ±.01	1.41 ±.03	1.45 ±.03	1.42 ±.02
a — inact.	0.00	0.87 ±.04	0.84 ±.04	0.86 ±.03	0.90 ±.01	0.85 ±.04	0.88 ±.06
μ — active	11.34	7.66 ±.13	7.83 ±.19	8.06 ±.14	7.74 ±.35	7.78 ±.07	8.30 ±.25
μ — inact.	—	0.82 ±.16	0.60 ±.67	0.90 ±.26	0.59 ±.34	0.21 ±.29	0.83 ±.30
σ — active	3.53	3.23 ±.06	3.23 ±.05	3.28 ±.06	3.24 ±.07	3.28 ±.09	3.22 ±.03
σ — inact.	—	1.81 ±.02	1.83 ±.04	1.85 ±.05	1.82 ±.06	1.84 ±.06	1.81 ±.04
distribution width — std over GT-active pixels							
a — slot 0	1.48	1.28 ±.03	1.30 ±.05	1.29 ±.02	1.27 ±.03	1.31 ±.03	1.36 ±.02
μ — slot 0	3.30	2.33 ±.07	2.30 ±.06	2.28 ±.03	2.33 ±.07	2.37 ±.04	2.56 ±.06
σ — slot 0	2.84	1.15 ±.04	1.22 ±.04	1.20 ±.05	1.17 ±.02	1.20 ±.02	1.23 ±.03
a — slot 1	1.45	1.28 ±.02	1.28 ±.03	1.28 ±.00	1.23 ±.06	1.27 ±.04	1.29 ±.01
μ — slot 1	16.24	8.85 ±.09	8.88 ±.49	8.88 ±.27	8.66 ±.29	8.96 ±.25	9.98 ±.31
σ — slot 1	4.01	1.72 ±.03	1.71 ±.06	1.74 ±.03	1.69 ±.07	1.74 ±.08	1.77 ±.04

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks). Column = parameter:existence split of the parity budget. Five seeded replicas per arm, cells mean ± seed std; means conditioned on the slot's GT activity mask. single = the benchmark UNet-skip · set-pred · param-L1 single trunk (31.35 M). **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; every activity and statistics row moves by less than combined seed noise across the five splits — a 3.1M existence arm calibrates like a 15.6M one. The single trunk keeps the widest predicted spreads, closest to the GT on every width row — the split arms compress their distributions slightly more. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; *a* in linear amplitude.

The ratio at $K=5$ — a tenth of the budget detects exactly as half does

metric	parameter : existence split				
	50–50	60–40	70–30	80–20	90–10
<i>training (validation)</i>					
val loss ↓	0.130 ±.000	0.129 ±.000	0.129 ±.000	0.129 ±.000	0.129 ±.000
<i>reconstruction (curve)</i>					
curve R^2 ↑	0.679 ±.010	0.682 ±.008	0.685 ±.004	0.675 ±.014	0.680 ±.010
curve MAE ↓	0.046 ±.000	0.046 ±.001	0.046 ±.000	0.046 ±.001	0.046 ±.001
PSNR [dB] ↑	51.0 ±.1	51.1 ±.1	51.1 ±.1	51.0 ±.2	51.0 ±.1
SSIM elev ↑	0.961 ±.000	0.961 ±.001	0.961 ±.000	0.961 ±.000	0.961 ±.000
SSIM range ↑	0.940 ±.001	0.940 ±.001	0.940 ±.001	0.941 ±.001	0.940 ±.001
SSIM azim ↑	0.954 ±.000	0.954 ±.001	0.954 ±.001	0.955 ±.000	0.954 ±.000
profile cos med ↑	0.922 ±.011	0.927 ±.002	0.929 ±.005	0.927 ±.009	0.924 ±.006
<i>detection (matched)</i>					
precision ↑	0.917 ±.006	0.908 ±.006	0.911 ±.007	0.915 ±.005	0.912 ±.007
recall ↑	0.666 ±.008	0.666 ±.018	0.668 ±.015	0.683 ±.009	0.669 ±.013
matched F1 ↑	0.772 ±.007	0.768 ±.012	0.771 ±.010	0.783 ±.006	0.772 ±.009
count exact ↑	0.632 ±.012	0.626 ±.025	0.630 ±.019	0.651 ±.011	0.634 ±.018
under ↓	0.329 ±.011	0.331 ±.025	0.327 ±.019	0.309 ±.011	0.326 ±.018
over ↓	0.039 ±.003	0.043 ±.002	0.042 ±.003	0.040 ±.002	0.040 ±.003
peak err med ↓	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00
peak err p95 ↓	14.1 ±.0	14.1 ±.0	14.1 ±.0	14.1 ±.0	14.1 ±.0

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks). Column = parameter:existence split of the ≈ 31.19 M parity budget (four-level ladders, only width moves); the 50–50 arm is the $K=5$ routing block's all-channels reference arm. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px, $K_{\max}=5$ labels. **bold** = best in row, marked only where best and worst differ by more than 5%. Nothing moves: the curve rows are flat, the detection edges reach at most the rim of combined seed noise (80–20 recall 0.683 vs 0.666), and the $K=5$ under-count (0.31–0.33) stands at every split — the conservative gate is not capacity-bound. Peak err med/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

The $K=5$ ratio by scatterer count — the recall ladder is split-blind

metric	parameter : existence split				
	50–50	60–40	70–30	80–20	90–10
precision $\uparrow$	0.917 $\pm$.006	0.908 $\pm$.006	0.911 $\pm$.007	0.915 $\pm$.005	0.912 $\pm$.007
$k=1$	0.936 $\pm$.005	0.930 $\pm$.004	0.929 $\pm$.004	0.937 $\pm$.002	0.936 $\pm$.005
$k=2$	0.912 $\pm$.008	0.905 $\pm$.006	0.910 $\pm$.009	0.911 $\pm$.007	0.908 $\pm$.009
$k=3$	0.869 $\pm$.010	0.856 $\pm$.017	0.864 $\pm$.013	0.863 $\pm$.010	0.855 $\pm$.018
$k=4$	0.819 $\pm$.013	0.792 $\pm$.030	0.807 $\pm$.018	0.799 $\pm$.018	0.790 $\pm$.030
$k=5$	0.794 $\pm$.015	0.756 $\pm$.034	0.774 $\pm$.021	0.763 $\pm$.027	0.751 $\pm$.037
recall $\uparrow$	0.666 $\pm$.008	0.666 $\pm$.018	0.668 $\pm$.015	0.683 $\pm$.009	0.669 $\pm$.013
$k=1$	0.757 $\pm$.014	0.750 $\pm$.032	0.755 $\pm$.024	0.783 $\pm$.013	0.758 $\pm$.024
$k=2$	0.701 $\pm$.008	0.710 $\pm$.008	0.710 $\pm$.005	0.711 $\pm$.005	0.707 $\pm$.008
$k=3$	0.508 $\pm$.005	0.513 $\pm$.003	0.511 $\pm$.006	0.511 $\pm$.003	0.511 $\pm$.004
$k=4$	0.358 $\pm$.008	0.363 $\pm$.002	0.358 $\pm$.010	0.360 $\pm$.004	0.362 $\pm$.010
$k=5$	0.239 $\pm$.008	0.248 $\pm$.004	0.241 $\pm$.014	0.245 $\pm$.005	0.248 $\pm$.014
count acc pred k $\uparrow$					
pred $k=1$	0.885 $\pm$.005	0.887 $\pm$.007	0.886 $\pm$.006	0.885 $\pm$.005	0.884 $\pm$.004
pred $k=2$	0.496 $\pm$.005	0.493 $\pm$.011	0.490 $\pm$.006	0.494 $\pm$.008	0.501 $\pm$.004
pred $k=3$	0.252 $\pm$.012	0.235 $\pm$.023	0.242 $\pm$.007	0.247 $\pm$.008	0.244 $\pm$.007
pred $k=4$	0.198 $\pm$.014	0.202 $\pm$.008	0.204 $\pm$.020	0.207 $\pm$.005	0.196 $\pm$.011
pred $k=5$	0.198 $\pm$.039	0.207 $\pm$.010	0.182 $\pm$.016	0.208 $\pm$.027	0.217 $\pm$.020

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks). Column = parameter: existence split of the ≈ 31.19 M parity budget (four-level ladders, only width moves); the 50–50 arm is the $K=5$ routing block's all-channels reference arm. All precision / recall rows are matched (Hungarian pairing); cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel. **bold** = best in row, marked only where best and worst differ by more than 5%. Every recall rung moves by less than combined seed noise across the splits; 50–50 keeps a within-noise precision edge on the deep rungs. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k .

The $K=5$ ratio components — the freed capacity buys nothing the noise can see

metric	parameter : existence split				
	50–50	60–40	70–30	80–20	90–10
μ MAE ↓	1.16 ±.05	1.20 ±.04	1.16 ±.06	1.18 ±.03	1.20 ±.06
$k=1$	0.48 ±.03	0.46 ±.01	0.46 ±.01	0.47 ±.03	0.47 ±.02
$k=2$	1.81 ±.03	1.83 ±.06	1.80 ±.04	1.84 ±.05	1.83 ±.05
$k=3$	2.55 ±.18	2.57 ±.17	2.47 ±.20	2.54 ±.10	2.60 ±.13
$k=4$	3.26 ±.26	3.54 ±.42	3.27 ±.30	3.42 ±.19	3.55 ±.40
$k=5$	3.52 ±.23	4.05 ±.54	3.77 ±.34	3.92 ±.41	4.10 ±.56
σ MAE ↓	0.60 ±.01	0.61 ±.02	0.61 ±.02	0.60 ±.01	0.61 ±.02
$k=1$	0.27 ±.01	0.26 ±.01	0.28 ±.00	0.27 ±.00	0.27 ±.01
$k=2$	0.88 ±.01	0.89 ±.02	0.89 ±.01	0.89 ±.02	0.88 ±.01
$k=3$	1.31 ±.02	1.34 ±.02	1.33 ±.03	1.34 ±.04	1.33 ±.03
$k=4$	1.65 ±.02	1.69 ±.02	1.68 ±.04	1.69 ±.03	1.70 ±.05
$k=5$	1.83 ±.02	1.86 ±.03	1.87 ±.06	1.87 ±.04	1.88 ±.08
a MAE ↓	0.212 ±.005	0.204 ±.015	0.206 ±.007	0.202 ±.015	0.213 ±.005
$k=1$	0.076 ±.004	0.074 ±.006	0.076 ±.004	0.071 ±.007	0.079 ±.004
$k=2$	0.387 ±.013	0.364 ±.019	0.375 ±.021	0.376 ±.022	0.387 ±.012
$k=3$	0.464 ±.013	0.438 ±.029	0.445 ±.013	0.446 ±.030	0.455 ±.011
$k=4$	0.498 ±.016	0.466 ±.040	0.463 ±.012	0.478 ±.038	0.479 ±.012
$k=5$	0.498 ±.023	0.461 ±.044	0.450 ±.021	0.468 ±.043	0.466 ±.016

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks). Column = parameter: existence split of the ≈ 31.19 M parity budget (four-level ladders, only width moves); the 50–50 arm is the $K=5$ routing block's all-channels reference arm. Cells mean ± seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only. **bold** = best in row, marked only where best and worst differ by more than 5%. Every $\mu/\sigma/a$ gap sits inside combined seed noise — at $K=5$ moving capacity into the parameter arm buys no localisation, where the same move cleared noise at $K=2$. μ, σ in elevation units; a in linear amplitude.

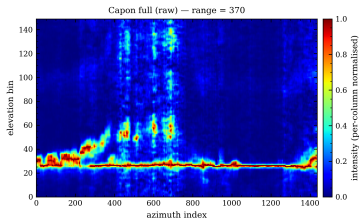
The $K=5$ ratio, predicted statistics vs the GT — the under-fired gate is not capacity-starved

		parameter : existence split				
mean	GT	50–50	60–40	70–30	80–20	90–10
<i>predicted active fraction — share of pixels where the slot fires</i>						
slot 0	1.000	0.762 ± .011	0.763 ± .027	0.771 ± .014	0.779 ± .021	0.755 ± .018
slot 1	0.239	0.160 ± .005	0.165 ± .009	0.167 ± .009	0.183 ± .016	0.177 ± .010
slot 2	0.085	0.037 ± .006	0.043 ± .011	0.049 ± .009	0.040 ± .015	0.043 ± .009
slot 3	0.035	0.009 ± .002	0.016 ± .010	0.010 ± .002	0.017 ± .016	0.020 ± .014
slot 4	0.015	0.030 ± .009	0.019 ± .010	0.010 ± .009	0.006 ± .005	0.012 ± .010
<i>a — predicted mean over GT-active pixels</i>						
slot 0	0.84	0.94 ± .02	0.97 ± .04	0.95 ± .04	0.94 ± .03	0.92 ± .03
slot 1	1.67	1.82 ± .04	1.80 ± .04	1.79 ± .07	1.79 ± .05	1.79 ± .02
slot 2	1.41	1.81 ± .03	1.75 ± .10	1.77 ± .11	1.75 ± .07	1.68 ± .08
slot 3	0.97	1.29 ± .06	1.11 ± .22	1.16 ± .15	1.08 ± .19	1.09 ± .15
slot 4	0.66	0.65 ± .10	0.49 ± .08	0.56 ± .09	0.46 ± .08	0.59 ± .08
<i>a — predicted mean over GT-inactive pixels (leakage)</i>						
slot 1	0.00	0.79 ± .86	1.01 ± .84	1.56 ± .81	1.15 ± .86	0.78 ± .58
slot 2	0.00	0.69 ± .60	0.61 ± .51	1.21 ± .56	0.52 ± .63	1.16 ± .77
slot 3	0.00	0.33 ± .19	0.66 ± .52	0.29 ± .12	0.56 ± .61	0.87 ± .59
slot 4	0.00	1.53 ± .16	1.17 ± .78	0.43 ± .46	0.20 ± .03	0.58 ± .60
<i>distribution width — std over GT-active pixels</i>						
a — slot 0	1.59	1.48 ± .01	1.53 ± .07	1.50 ± .05	1.52 ± .06	1.48 ± .04
μ — slot 0	3.02	2.41 ± .06	2.52 ± .07	2.44 ± .07	2.44 ± .08	2.52 ± .03
σ — slot 0	1.72	1.03 ± .04	1.07 ± .06	1.06 ± .08	1.03 ± .08	1.06 ± .04
a — slot 1	1.60	1.38 ± .03	1.39 ± .04	1.36 ± .04	1.42 ± .04	1.40 ± .03
μ — slot 1	11.09	8.96 ± .40	8.74 ± .12	8.64 ± .41	8.85 ± .18	8.85 ± .26
σ — slot 1	2.30	1.40 ± .02	1.40 ± .04	1.40 ± .05	1.42 ± .04	1.44 ± .04

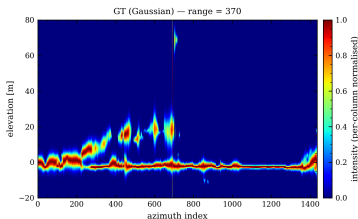
runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · param-L1 · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks). Column = parameter: existence split of the ≈ 31.19 M parity budget (four-level ladders, only width moves); the 50–50 arm is the $K=5$ routing block's all-channels reference arm. Five seeded replicas per arm, cells mean ± seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%. Slot 0 fires on 0.76–0.78 of pixels at every split — a 3.1 M existence arm under-fires exactly like a 15.6 M one, so the conservative $K=5$ gate is not starved for capacity. Width rows: std of the prediction over GT-active pixels vs the GT's own std, shown for slots 0–1 (tail slots in the report). μ , σ in elevation units; a in linear amplitude.

What the 90–10 model reconstructs — one range cut, end to end

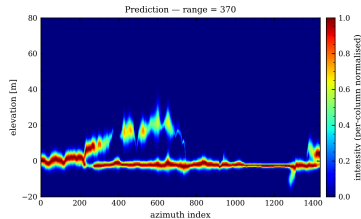
full-tomogram



GT (Gaussian)



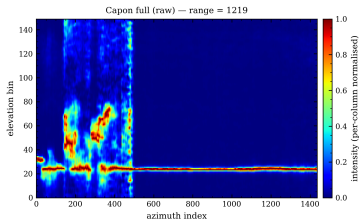
90–10 prediction



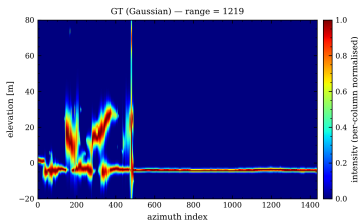
One constant-range cut ($rg = 370$) through the held-out cube, azimuth on the horizontal axis, elevation on the vertical, every column normalised to its own peak. **Left:** the raw Capon tomogram the labels were fitted from, indexed in elevation bins on the full inversion grid — its vertical axis is not on the metre scale of the other two. **Middle:** the $K=2$ Gaussian mixture re-synthesised from the GT parameters, $z \in [-20, 80]$ m. **Right:** the same mixture re-synthesised from the 90–10 arm's predicted parameters, one seed. Ground return, the canopy ramp at azimuth 220–400 and the elevated returns at 420–720 all land where the label puts them; the isolated 70 m spike near azimuth 700 does not survive into the prediction.

The 90–10 cube — canopy against bare ground

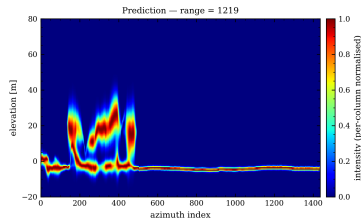
full-tomogram



GT (Gaussian)



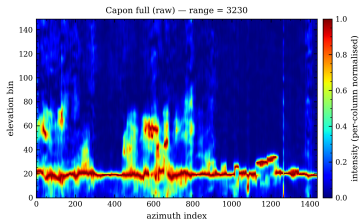
90–10 prediction



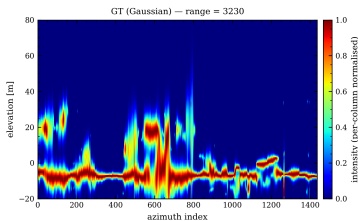
Constant-range cut at $rg = 1219$, per-column normalised. The tomogram splits the scene before any fit: canopy clutter out to azimuth ≈ 500 , then a single clean ground line to the swath edge. Label and prediction both follow that split, and the prediction holds the ground line at ≈ -4 m across the whole flat half. Its vegetated half comes out as broad smooth lobes where the label is filamentary — the patch CNN regularises across neighbouring azimuths. The single-column GT streak at azimuth ≈ 480 sits on the tomogram's bright edge column but does not reach the prediction. Tomogram in elevation bins, the two mixture panels in metres.

The 90–10 cube — layered canopy over a continuous ground return

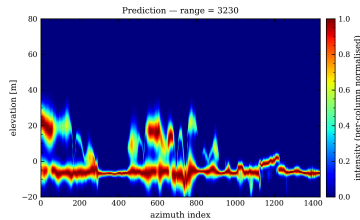
full-tomogram



GT (Gaussian)



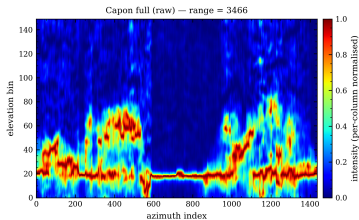
90–10 prediction



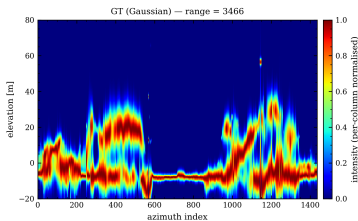
Constant-range cut at $rg = 3230$, per-column normalised. Two layers over roughly the first 850 azimuths in all three panels: a ground return near -6 m under vegetation at 15 – 25 m. The prediction keeps the separation, and the raised patch at azimuth 1130 – 1220 , where the return lifts to about 0 m before dropping back, shows in the tomogram, the label and the prediction alike. The thin GT column at azimuth ≈ 790 reaching past 60 m is cut short in the prediction — isolated single-column towers are the failure mode these cuts keep showing. Tomogram in elevation bins, the two mixture panels in metres.

The 90–10 cube — dense canopy with a bare corridor

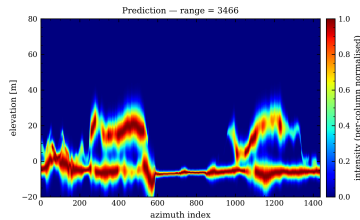
full-tomogram



GT (Gaussian)



90–10 prediction



Constant-range cut at $rg = 3466$, per-column normalised: the densest of the four. Vegetation from azimuth 0 to ≈ 580 and again from ≈ 950 out to the swath edge, with a bare corridor between them where only the ground return near -7 m survives; the tomogram carries the same corridor. The prediction puts both blocks and the corridor in place and holds the sharp canopy wall at azimuth ≈ 580 . Two costs: it picks the canopy back up at azimuth ≈ 985 against the label's ≈ 955 , and the ≈ 60 m point return near azimuth 1145, visible in the tomogram as well as in the label, is missing. Tomogram in elevation bins, the two mixture panels in metres.

Pairing the objective — param-L1 plus a curve term

$$\mathcal{L}(w) = (\mathcal{L}_{\text{param-L1}} + w \mathcal{L}_{\text{curve}}) / (1 + w)$$



1 baseline + 15 pair arms per K , five seeds each, at $K=2$ and $K=5$ = 160 runs

The ratio block left the $K=5$ gate under-firing at every split — more existence capacity did not wake it. This experiment changes the objective instead: keep param-L1 and add a synthesized-curve term.

- **Two signals, one loss:** param-L1 reaches a slot only through its matched pair; the curve term re-synthesizes the profile from *all* slots and is permutation-invariant — its gradient touches every slot, matched or not.
- **Magnitude or shape:** L1 and Charbonnier price the synthesized profile's magnitude; cosine normalizes both curves first and prices shape alone.
- **A normalized pair:** the two terms are averaged, so the validation loss changes meaning with w — arms compare on held-out metrics only, never on the loss.
- **The question:** if the conservative gate is starved of a profile-level error signal rather than capacity, a small w should move detection at $K=5$ and leave $K=2$ alone.

The pair sweep at $K=2$ — the curve term rides free, cosine shifts the operating point

objective	reconstruction		detection (matched)						
	curve $R^2 \uparrow$	cos med $\uparrow$	prec. $\uparrow$	recall $\uparrow$	F1 $\uparrow$	exact $\uparrow$	under $\downarrow$	over $\downarrow$	peak p95 $\downarrow$
param-L1 alone	0.748 $\pm$.010	0.969 $\pm$.001	0.905 $\pm$.003	0.891 $\pm$.003	0.898 $\pm$.002	0.910 $\pm$.004	0.049 $\pm$.002	0.040 $\pm$.004	5.4 $\pm$.0
+ <i>L1 curve</i>									
$w=0.01$	0.743 $\pm$.011	0.967 $\pm$.003	0.905 $\pm$.001	0.890 $\pm$.001	0.898 $\pm$.001	0.909 $\pm$.003	0.050 $\pm$.002	0.041 $\pm$.002	5.4 $\pm$.0
$w=0.05$	0.749 $\pm$.008	0.968 $\pm$.002	0.906 $\pm$.003	0.889 $\pm$.001	0.897 $\pm$.002	0.910 $\pm$.004	0.051 $\pm$.002	0.040 $\pm$.004	5.4 $\pm$.0
$w=0.25$	0.745 $\pm$.012	0.967 $\pm$.001	0.905 $\pm$.004	0.889 $\pm$.004	0.897 $\pm$.003	0.908 $\pm$.005	0.051 $\pm$.004	0.041 $\pm$.005	5.6 $\pm$.4
$w=0.5$	0.757 $\pm$.011	0.969 $\pm$.003	0.905 $\pm$.003	0.890 $\pm$.003	0.897 $\pm$.001	0.910 $\pm$.003	0.050 $\pm$.004	0.040 $\pm$.002	5.5 $\pm$.3
$w=1$	0.753 $\pm$.010	0.967 $\pm$.004	0.905 $\pm$.004	0.889 $\pm$.003	0.897 $\pm$.003	0.909 $\pm$.007	0.051 $\pm$.004	0.040 $\pm$.006	5.6 $\pm$.4
+ <i>Charbonnier curve</i>									
$w=0.01$	0.745 $\pm$.007	0.967 $\pm$.003	0.905 $\pm$.004	0.889 $\pm$.002	0.897 $\pm$.002	0.909 $\pm$.005	0.050 $\pm$.001	0.040 $\pm$.005	5.6 $\pm$.4
$w=0.05$	0.752 $\pm$.008	0.966 $\pm$.005	0.906 $\pm$.002	0.889 $\pm$.001	0.897 $\pm$.001	0.911 $\pm$.004	0.050 $\pm$.001	0.038 $\pm$.004	5.4 $\pm$.0
$w=0.25$	0.749 $\pm$.012	0.968 $\pm$.003	0.907 $\pm$.003	0.889 $\pm$.003	0.898 $\pm$.002	0.909 $\pm$.006	0.052 $\pm$.004	0.039 $\pm$.004	5.6 $\pm$.6
$w=0.5$	0.749 $\pm$.014	0.969 $\pm$.002	0.906 $\pm$.008	0.888 $\pm$.004	0.897 $\pm$.005	0.905 $\pm$.013	0.053 $\pm$.005	0.041 $\pm$.011	5.6 $\pm$.4
$w=1$	0.749 $\pm$.007	0.966 $\pm$.007	0.906 $\pm$.006	0.888 $\pm$.002	0.897 $\pm$.002	0.909 $\pm$.005	0.052 $\pm$.005	0.039 $\pm$.006	5.6 $\pm$.4
+ <i>cosine curve</i>									
$w=0.01$	0.742 $\pm$.005	0.968 $\pm$.002	0.900 $\pm$.005	0.898 $\pm$.002	0.899 $\pm$.002	0.913 $\pm$.005	0.044 $\pm$.002	0.043 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.745 $\pm$.010	0.966 $\pm$.005	0.900 $\pm$.005	0.898 $\pm$.001	0.899 $\pm$.002	0.916 $\pm$.005	0.043 $\pm$.004	0.041 $\pm$.005	4.7 $\pm$.0
$w=0.25$	0.746 $\pm$.015	0.966 $\pm$.004	0.894 $\pm$.003	0.900 $\pm$.002	0.897 $\pm$.002	0.909 $\pm$.006	0.042 $\pm$.003	0.049 $\pm$.004	4.7 $\pm$.0
$w=0.5$	0.754 $\pm$.008	0.968 $\pm$.002	0.891 $\pm$.008	0.902 $\pm$.002	0.897 $\pm$.004	0.905 $\pm$.010	0.040 $\pm$.002	0.055 $\pm$.011	4.7 $\pm$.0
$w=1$	0.749 $\pm$.003	0.966 $\pm$.002	0.884 $\pm$.010	0.902 $\pm$.002	0.893 $\pm$.005	0.900 $\pm$.011	0.037 $\pm$.003	0.064 $\pm$.013	4.7 $\pm$.0

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twin ladders (≈ 31.19 M). Row = objective ($\mathcal{L}_{\text{param-L1}} + w \mathcal{L}_{\text{curve}})/(1 + w)$. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. Validation losses are omitted — every arm optimizes a different objective, so they do not compare. Curve MAE, PSNR and three-axis SSIM are flat over all sixteen arms (0.036–0.037, 52.6–52.8 dB, elev/range/azim 0.967–0.968 / 0.951–0.952 / 0.961–0.962). **bold** = best in column, marked only where best and worst differ by more than 5%. **Green** and **red** sit on one row: full-weight cosine trades misses for false alarms (under 0.049 $\rightarrow$ 0.037, over 0.040 $\rightarrow$ 0.064) at flat F1 — an operating-point shift, not extra detection. Every other column stays inside combined seed noise: at $K=2$ the pair term rides along free. cos med: median per-pixel cosine to the GT profile; peak p95: 95th percentile of the per-pixel peak error.

The $K=2$ pair by scatterer count — wider firing buys the second scatterer, position pays

metric	paired curve term					
	none	L1 $w=1$	Charb. $w=1$	cos. $w=0.01$	cos. $w=0.5$	cos. $w=1$
precision $\uparrow$	0.905 $\pm$.003	0.905 $\pm$.004	0.906 $\pm$.006	0.900 $\pm$.005	0.891 $\pm$.008	0.884 $\pm$.010
$k=1$	0.947 $\pm$.005	0.948 $\pm$.007	0.950 $\pm$.008	0.944 $\pm$.006	0.930 $\pm$.012	0.920 $\pm$.015
$k=2$	0.834 $\pm$.002	0.833 $\pm$.002	0.833 $\pm$.004	0.828 $\pm$.003	0.827 $\pm$.004	0.823 $\pm$.003
recall $\uparrow$	0.891 $\pm$.003	0.889 $\pm$.003	0.888 $\pm$.002	0.898 $\pm$.002	0.902 $\pm$.002	0.902 $\pm$.002
$k=1$	0.995 $\pm$.002	0.995 $\pm$.001	0.995 $\pm$.001	0.998 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000
$k=2$	0.742 $\pm$.006	0.737 $\pm$.005	0.735 $\pm$.006	0.755 $\pm$.003	0.764 $\pm$.005	0.765 $\pm$.004
count acc pred k						
pred $k=1$ $\uparrow$	0.952 $\pm$.003	0.950 $\pm$.004	0.950 $\pm$.005	0.944 $\pm$.003	0.945 $\pm$.002	0.949 $\pm$.003
pred $k=2$ $\uparrow$	0.841 $\pm$.013	0.841 $\pm$.019	0.846 $\pm$.020	0.835 $\pm$.016	0.802 $\pm$.029	0.780 $\pm$.032
μ MAE $\downarrow$	1.59 $\pm$.02	1.61 $\pm$.01	1.60 $\pm$.02	1.68 $\pm$.04	1.69 $\pm$.04	1.73 $\pm$.04
$k=1$	0.31 $\pm$.01	0.33 $\pm$.01	0.33 $\pm$.03	0.34 $\pm$.01	0.35 $\pm$.01	0.36 $\pm$.01
$k=2$	3.63 $\pm$.05	3.66 $\pm$.03	3.65 $\pm$.06	3.77 $\pm$.07	3.76 $\pm$.09	3.84 $\pm$.09
σ MAE $\downarrow$	0.690 $\pm$.008	0.690 $\pm$.008	0.686 $\pm$.005	0.749 $\pm$.024	0.765 $\pm$.005	0.783 $\pm$.013
$k=1$	0.185 $\pm$.005	0.179 $\pm$.006	0.180 $\pm$.007	0.199 $\pm$.005	0.199 $\pm$.003	0.200 $\pm$.004
$k=2$	1.494 $\pm$.021	1.506 $\pm$.014	1.497 $\pm$.008	1.605 $\pm$.048	1.635 $\pm$.017	1.675 $\pm$.026
a MAE $\downarrow$	0.177 $\pm$.002	0.188 $\pm$.006	0.190 $\pm$.007	0.176 $\pm$.003	0.182 $\pm$.003	0.186 $\pm$.003
$k=1$	0.056 $\pm$.003	0.059 $\pm$.005	0.060 $\pm$.005	0.057 $\pm$.004	0.059 $\pm$.002	0.060 $\pm$.003
$k=2$	0.371 $\pm$.002	0.396 $\pm$.009	0.399 $\pm$.010	0.363 $\pm$.003	0.372 $\pm$.004	0.379 $\pm$.004

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = paired curve term at its weight w ; none = param-L1 alone. All rows matched (Hungarian pairing); cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only. **bold** = best in row, marked only where best and worst differ by more than 5%. **Green** = the cosine arms' two-scatterer recall (0.742 $\rightarrow$ 0.765, $\approx 3 \times$ combined seed noise); **red** = what the wider gate costs: μ and σ drift up with cosine weight (μ 1.59 $\rightarrow$ 1.73, σ 0.690 $\rightarrow$ 0.783, both past noise) while $k=1$ precision slips inside the gate (0.947 $\rightarrow$ 0.920). The magnitude pairs move nothing anywhere. Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

The $K=2$ pair, predicted statistics — the gate fires wider, leakage thins

		paired curve term					
mean	GT	none	L1 $w=1$	Charb. $w=1$	cos. $w=0.01$	cos. $w=0.5$	cos. $w=1$
<i>slot 0 — strongest scatterer; GT-active on every pixel</i>							
active frac	1.000	0.972 $\pm$.003	0.970 $\pm$.002	0.970 $\pm$.002	0.983 $\pm$.002	0.988 $\pm$.001	0.988 $\pm$.001
a — active	0.77	0.74 $\pm$.01	0.72 $\pm$.01	0.72 $\pm$.01	0.73 $\pm$.01	0.71 $\pm$.01	0.71 $\pm$.01
μ — active	-4.89	-4.58 $\pm$.05	-4.58 $\pm$.10	-4.57 $\pm$.12	-4.61 $\pm$.07	-4.61 $\pm$.07	-4.63 $\pm$.07
σ — active	1.76	1.38 $\pm$.03	1.37 $\pm$.02	1.36 $\pm$.05	1.40 $\pm$.01	1.48 $\pm$.02	1.49 $\pm$.01
<i>slot 1 — second scatterer; GT-active on 26% of pixels</i>							
active frac	0.260	0.268 $\pm$.004	0.267 $\pm$.006	0.265 $\pm$.008	0.274 $\pm$.006	0.288 $\pm$.011	0.299 $\pm$.015
a — active	1.43	1.45 $\pm$.03	1.36 $\pm$.03	1.35 $\pm$.04	1.42 $\pm$.03	1.39 $\pm$.03	1.40 $\pm$.02
a — inact.	0.00	0.85 $\pm$.04	0.74 $\pm$.10	0.78 $\pm$.10	0.84 $\pm$.04	0.63 $\pm$.08	0.57 $\pm$.07
μ — active	11.34	7.78 $\pm$.07	7.44 $\pm$.36	7.39 $\pm$.36	7.75 $\pm$.28	7.33 $\pm$.23	7.30 $\pm$.19
μ — inact.	—	0.21 $\pm$.29	0.45 $\pm$.67	0.22 $\pm$.95	0.10 $\pm$.59	-0.80 $\pm$.74	-1.08 $\pm$ 1.08
σ — active	3.53	3.28 $\pm$.09	3.29 $\pm$.13	3.24 $\pm$.13	3.24 $\pm$.07	3.33 $\pm$.05	3.47 $\pm$.03
σ — inact.	—	1.84 $\pm$.06	1.78 $\pm$.06	1.79 $\pm$.05	1.78 $\pm$.04	1.79 $\pm$.07	1.78 $\pm$.09
<i>distribution width — std over GT-active pixels</i>							
a — slot 0	1.48	1.31 $\pm$.03	1.31 $\pm$.03	1.29 $\pm$.04	1.29 $\pm$.02	1.26 $\pm$.03	1.26 $\pm$.03
μ — slot 0	3.30	2.37 $\pm$.04	2.39 $\pm$.06	2.40 $\pm$.06	2.44 $\pm$.05	2.44 $\pm$.05	2.44 $\pm$.08
σ — slot 0	2.84	1.20 $\pm$.02	1.15 $\pm$.08	1.16 $\pm$.08	1.25 $\pm$.05	1.38 $\pm$.06	1.38 $\pm$.03
a — slot 1	1.45	1.27 $\pm$.04	1.21 $\pm$.03	1.21 $\pm$.04	1.27 $\pm$.05	1.29 $\pm$.03	1.33 $\pm$.02
μ — slot 1	16.24	8.96 $\pm$.25	8.86 $\pm$.33	8.83 $\pm$.30	9.33 $\pm$.40	8.99 $\pm$.43	9.05 $\pm$.51
σ — slot 1	4.01	1.74 $\pm$.08	1.81 $\pm$.12	1.79 $\pm$.13	1.74 $\pm$.07	1.80 $\pm$.03	1.99 $\pm$.13

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = paired curve term at its weight w ; none = param-L1 alone. Five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%. The cosine gate fires wider on both slots (slot 0: 0.972 $\rightarrow$ 0.988; slot 1: 0.268 $\rightarrow$ 0.299 against GT 0.260 — past the GT into over-fire), thins the inactive-amplitude leakage (0.85 $\rightarrow$ 0.57) and re-widens the compressed σ statistics toward the GT (slot-0 σ width 1.20 $\rightarrow$ 1.38 vs 2.84) — the shape term pushes the predicted distributions outward at $K=2$ too, it just finds nothing new to detect. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; a in linear amplitude.

The pair sweep at $K=5$ — a shape term at trace weight wakes the gate

objective	reconstruction		detection (matched)						
	curve $R^2 \uparrow$	cos med $\uparrow$	prec. $\uparrow$	recall $\uparrow$	F1 $\uparrow$	exact $\uparrow$	under $\downarrow$	over $\downarrow$	peak p95 $\downarrow$
param-L1 alone	0.680 $\pm$.010	0.924 $\pm$.006	0.912 $\pm$.007	0.669 $\pm$.013	0.772 $\pm$.009	0.634 $\pm$.018	0.326 $\pm$.018	0.040 $\pm$.003	14.1 $\pm$ 0
+ <i>L1 curve</i>									
$w=0.01$	0.680 $\pm$.012	0.919 $\pm$.008	0.906 $\pm$.007	0.660 $\pm$.019	0.764 $\pm$.014	0.618 $\pm$.026	0.339 $\pm$.025	0.043 $\pm$.002	14.1 $\pm$ 0
$w=0.05$	0.684 $\pm$.006	0.922 $\pm$.007	0.913 $\pm$.005	0.657 $\pm$.017	0.764 $\pm$.013	0.618 $\pm$.024	0.342 $\pm$.022	0.039 $\pm$.002	14.1 $\pm$ 0
$w=0.25$	0.700 $\pm$.007	0.928 $\pm$.005	0.910 $\pm$.005	0.680 $\pm$.009	0.779 $\pm$.007	0.644 $\pm$.010	0.312 $\pm$.011	0.044 $\pm$.002	14.1 $\pm$ 0
$w=0.5$	0.702 $\pm$.008	0.940 $\pm$.005	0.914 $\pm$.004	0.715 $\pm$.019	0.802 $\pm$.013	0.687 $\pm$.027	0.268 $\pm$.027	0.045 $\pm$.002	14.0 $\pm$ 3
$w=1$	0.702 $\pm$.020	0.950 $\pm$.005	0.917 $\pm$.009	0.801 $\pm$.006	0.855 $\pm$.003	0.793 $\pm$.008	0.154 $\pm$.007	0.053 $\pm$.010	6.7 $\pm$ 7
+ <i>Charbonnier curve</i>									
$w=0.01$	0.682 $\pm$.014	0.924 $\pm$.008	0.912 $\pm$.004	0.665 $\pm$.016	0.769 $\pm$.011	0.629 $\pm$.021	0.331 $\pm$.021	0.041 $\pm$.001	14.1 $\pm$ 0
$w=0.05$	0.683 $\pm$.009	0.925 $\pm$.007	0.911 $\pm$.005	0.664 $\pm$.011	0.768 $\pm$.008	0.626 $\pm$.015	0.333 $\pm$.015	0.041 $\pm$.002	14.1 $\pm$ 0
$w=0.25$	0.699 $\pm$.005	0.927 $\pm$.004	0.910 $\pm$.005	0.681 $\pm$.015	0.779 $\pm$.010	0.644 $\pm$.018	0.312 $\pm$.020	0.044 $\pm$.003	14.1 $\pm$ 0
$w=0.5$	0.706 $\pm$.004	0.937 $\pm$.008	0.917 $\pm$.004	0.713 $\pm$.022	0.802 $\pm$.015	0.685 $\pm$.033	0.271 $\pm$.032	0.044 $\pm$.002	14.0 $\pm$ 3
$w=1$	0.712 $\pm$.010	0.948 $\pm$.010	0.915 $\pm$.012	0.794 $\pm$.015	0.850 $\pm$.013	0.781 $\pm$.033	0.165 $\pm$.021	0.054 $\pm$.013	8.3 $\pm$ 3.0
+ <i>cosine curve</i>									
$w=0.01$	0.683 $\pm$.011	0.949 $\pm$.005	0.918 $\pm$.007	0.813 $\pm$.003	0.863 $\pm$.002	0.810 $\pm$.003	0.142 $\pm$.004	0.048 $\pm$.004	5.1 $\pm$ 4
$w=0.05$	0.685 $\pm$.011	0.952 $\pm$.004	0.906 $\pm$.001	0.820 $\pm$.004	0.861 $\pm$.003	0.805 $\pm$.007	0.134 $\pm$.005	0.061 $\pm$.004	4.9 $\pm$ 3
$w=0.25$	0.700 $\pm$.006	0.951 $\pm$.002	0.895 $\pm$.005	0.831 $\pm$.001	0.862 $\pm$.002	0.806 $\pm$.004	0.121 $\pm$.001	0.072 $\pm$.005	4.7 $\pm$ 0
$w=0.5$	0.704 $\pm$.005	0.953 $\pm$.002	0.883 $\pm$.010	0.837 $\pm$.003	0.859 $\pm$.004	0.802 $\pm$.007	0.113 $\pm$.003	0.085 $\pm$.010	4.7 $\pm$ 0
$w=1$	0.700 $\pm$.005	0.954 $\pm$.002	0.867 $\pm$.009	0.843 $\pm$.001	0.855 $\pm$.004	0.794 $\pm$.010	0.104 $\pm$.002	0.102 $\pm$.011	4.7 $\pm$ 0

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twin ladders. Row = objective ($\mathcal{L}_{\text{param-L1}} + w \mathcal{L}_{\text{curve}})/(1 + w)$. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, $K_{\text{max}}=5$ labels. Validation losses are omitted — every arm optimizes a different objective, so they do not compare. Curve MAE, PSNR and three-axis SSIM barely move (0.044–0.046, 51.0–51.5 dB, elev/range/azim 0.961–0.963 / 0.940–0.947 / 0.954–0.957); curve R^2 rises with any strong curve term (0.680 $\rightarrow$ 0.712, $\approx 3 \times$ seed noise, under the 5% bold gate). **bold** = best in column past 5%. **Green** = the trace-weight cosine arm: recall +0.14, count exact +0.18, under-count 0.33 $\rightarrow$ 0.14 — while holding the board's best precision. **Red** = the price of pushing w : full-weight cosine pays precision (0.867) and over-fires (0.102). The magnitude terms buy the same wake-up only at $w=1$, and less of it. cos med: median per-pixel cosine to the GT profile; peak p95: 95th percentile of the per-pixel peak error.

The $K=5$ pair by scatterer count — the missing scatterers were single

metric	paired curve term					
	none	L1 $w=1$	Charb. $w=1$	cos. $w=0.01$	cos. $w=0.5$	cos. $w=1$
precision $\uparrow$	0.912 $\pm$.007	0.917 $\pm$.009	0.915 $\pm$.012	0.918 $\pm$.007	0.883 $\pm$.010	0.867 $\pm$.009
$k=1$	0.936 $\pm$.005	0.934 $\pm$.013	0.931 $\pm$.019	0.940 $\pm$.005	0.901 $\pm$.011	0.883 $\pm$.012
$k=2$	0.908 $\pm$.009	0.912 $\pm$.006	0.913 $\pm$.006	0.910 $\pm$.008	0.888 $\pm$.009	0.877 $\pm$.006
$k=3$	0.855 $\pm$.018	0.863 $\pm$.008	0.866 $\pm$.009	0.860 $\pm$.017	0.834 $\pm$.009	0.824 $\pm$.002
$k=4$	0.790 $\pm$.030	0.808 $\pm$.014	0.812 $\pm$.014	0.792 $\pm$.027	0.754 $\pm$.015	0.742 $\pm$.005
$k=5$	0.751 $\pm$.037	0.779 $\pm$.018	0.781 $\pm$.018	0.743 $\pm$.034	0.697 $\pm$.021	0.679 $\pm$.018
recall $\uparrow$	0.669 $\pm$.013	0.801 $\pm$.006	0.794 $\pm$.015	0.813 $\pm$.003	0.837 $\pm$.003	0.843 $\pm$.001
$k=1$	0.759 $\pm$.024	0.975 $\pm$.006	0.962 $\pm$.029	0.995 $\pm$.001	0.997 $\pm$.000	0.997 $\pm$.000
$k=2$	0.708 $\pm$.008	0.749 $\pm$.013	0.748 $\pm$.009	0.746 $\pm$.005	0.813 $\pm$.007	0.832 $\pm$.004
$k=3$	0.511 $\pm$.004	0.529 $\pm$.007	0.530 $\pm$.003	0.528 $\pm$.004	0.565 $\pm$.004	0.576 $\pm$.002
$k=4$	0.362 $\pm$.010	0.371 $\pm$.007	0.373 $\pm$.005	0.380 $\pm$.007	0.406 $\pm$.005	0.413 $\pm$.003
$k=5$	0.246 $\pm$.015	0.249 $\pm$.009	0.251 $\pm$.008	0.274 $\pm$.010	0.306 $\pm$.010	0.308 $\pm$.005
count acc pred $k \uparrow$						
pred $k=1$	0.884 $\pm$.004	0.910 $\pm$.002	0.906 $\pm$.008	0.900 $\pm$.004	0.937 $\pm$.005	0.950 $\pm$.006
pred $k=2$	0.501 $\pm$.004	0.483 $\pm$.017	0.480 $\pm$.023	0.482 $\pm$.010	0.460 $\pm$.010	0.448 $\pm$.013
pred $k=3$	0.243 $\pm$.007	0.253 $\pm$.014	0.250 $\pm$.034	0.220 $\pm$.017	0.211 $\pm$.028	0.208 $\pm$.035
pred $k=4$	0.196 $\pm$.012	0.217 $\pm$.025	0.227 $\pm$.034	0.200 $\pm$.008	0.201 $\pm$.013	0.207 $\pm$.012
pred $k=5$	0.216 $\pm$.020	0.226 $\pm$.062	0.192 $\pm$.036	0.194 $\pm$.038	0.204 $\pm$.033	0.259 $\pm$.111

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = paired curve term at its weight w ; none = param-L1 alone. All precision / recall rows matched (Hungarian pairing); cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel. **bold** = best in row past 5%. **Green**: the baseline's misses were mostly single scatterers — $k=1$ recall 0.759 $\rightarrow$ 0.995 at trace-weight cosine (0.975/0.962 for the magnitude terms at $w=1$); the deep rungs move far less ($k=5$ 0.246 $\rightarrow$ 0.308, heavy cosine only). **Red** = the full-weight cosine precision price. Claimed counts ≥ 2 stay unreliable in every arm (pred-2 accuracy ≤ 0.50). Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k .

The $K=5$ pair components — magnitude pairs localise, heavy cosine pays position

metric	paired curve term					
	none	L1 $w=1$	Charb. $w=1$	cos. $w=0.01$	cos. $w=0.5$	cos. $w=1$
μ MAE $\downarrow$	1.21 $\pm$.07	1.10 $\pm$.04	1.10 $\pm$.04	1.13 $\pm$.05	1.33 $\pm$.06	1.42 $\pm$.06
$k=1$	0.47 $\pm$.02	0.45 $\pm$.02	0.46 $\pm$.03	0.46 $\pm$.02	0.46 $\pm$.01	0.47 $\pm$.01
$k=2$	1.83 $\pm$.05	1.84 $\pm$.08	1.82 $\pm$.03	1.92 $\pm$.07	2.07 $\pm$.11	2.14 $\pm$.11
$k=3$	2.60 $\pm$.13	2.81 $\pm$.14	2.71 $\pm$.08	2.67 $\pm$.14	3.33 $\pm$.21	3.69 $\pm$.19
$k=4$	3.56 $\pm$.40	3.50 $\pm$.25	3.38 $\pm$.16	3.60 $\pm$.30	4.42 $\pm$.27	4.88 $\pm$.21
$k=5$	4.11 $\pm$.56	3.71 $\pm$.30	3.64 $\pm$.30	4.17 $\pm$.48	4.96 $\pm$.33	5.43 $\pm$.46
σ MAE $\downarrow$	0.61 $\pm$.02	0.57 $\pm$.01	0.57 $\pm$.01	0.58 $\pm$.01	0.62 $\pm$.01	0.64 $\pm$.01
$k=1$	0.27 $\pm$.01	0.26 $\pm$.01	0.26 $\pm$.01	0.27 $\pm$.01	0.24 $\pm$.00	0.25 $\pm$.00
$k=2$	0.88 $\pm$.01	0.89 $\pm$.01	0.90 $\pm$.01	0.90 $\pm$.01	0.94 $\pm$.01	0.97 $\pm$.02
$k=3$	1.34 $\pm$.03	1.40 $\pm$.01	1.40 $\pm$.01	1.38 $\pm$.02	1.49 $\pm$.01	1.54 $\pm$.03
$k=4$	1.70 $\pm$.05	1.78 $\pm$.03	1.78 $\pm$.03	1.80 $\pm$.03	1.95 $\pm$.03	2.02 $\pm$.06
$k=5$	1.88 $\pm$.08	2.00 $\pm$.04	2.00 $\pm$.06	2.09 $\pm$.03	2.28 $\pm$.05	2.39 $\pm$.11
a MAE $\downarrow$	0.213 $\pm$.005	0.204 $\pm$.006	0.206 $\pm$.009	0.168 $\pm$.003	0.190 $\pm$.007	0.203 $\pm$.006
$k=1$	0.079 $\pm$.004	0.065 $\pm$.004	0.067 $\pm$.005	0.057 $\pm$.003	0.064 $\pm$.006	0.066 $\pm$.003
$k=2$	0.387 $\pm$.011	0.417 $\pm$.011	0.417 $\pm$.017	0.351 $\pm$.007	0.372 $\pm$.010	0.392 $\pm$.013
$k=3$	0.455 $\pm$.011	0.522 $\pm$.013	0.524 $\pm$.022	0.421 $\pm$.009	0.459 $\pm$.010	0.495 $\pm$.013
$k=4$	0.479 $\pm$.012	0.561 $\pm$.011	0.566 $\pm$.023	0.432 $\pm$.013	0.472 $\pm$.013	0.510 $\pm$.015
$k=5$	0.465 $\pm$.016	0.539 $\pm$.012	0.540 $\pm$.039	0.379 $\pm$.016	0.391 $\pm$.018	0.418 $\pm$.024

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = paired curve term at its weight w ; none = param-L1 alone. Cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only — an arm that recalls more matches a harder pool, so read this frame with the recall frame. **bold** = best in row past 5%. **Green** = the magnitude pairs cut μ despite matching far more pairs (μ 1.21 $\rightarrow$ 1.10, $\approx 2 \times$ noise), and trace-weight cosine takes amplitude on every rung (a MAE 0.213 $\rightarrow$ 0.168). **Red** = heavy cosine pays position: μ 1.42 overall, 2.60 $\rightarrow$ 3.69 at $k=3$ — the ladder steepens as w grows. μ , σ in elevation units; a in linear amplitude.

The $K=5$ pair, predicted statistics — slot 0 comes back on

		paired curve term					
mean	GT	none	L1 $w=1$	Charb. $w=1$	cos. $w=0.01$	cos. $w=0.5$	cos. $w=1$
<i>predicted active fraction — share of pixels where the slot fires</i>							
slot 0	1.000	0.756 $\pm$.018	0.930 $\pm$.011	0.921 $\pm$.022	0.948 $\pm$.007	0.964 $\pm$.003	0.967 $\pm$.001
slot 1	0.239	0.177 $\pm$.010	0.183 $\pm$.011	0.184 $\pm$.012	0.191 $\pm$.012	0.228 $\pm$.021	0.218 $\pm$.015
slot 2	0.085	0.043 $\pm$.009	0.049 $\pm$.018	0.048 $\pm$.016	0.049 $\pm$.017	0.073 $\pm$.024	0.084 $\pm$.035
slot 3	0.035	0.020 $\pm$.014	0.026 $\pm$.019	0.024 $\pm$.020	0.017 $\pm$.013	0.019 $\pm$.009	0.037 $\pm$.027
slot 4	0.015	0.012 $\pm$.010	0.014 $\pm$.013	0.015 $\pm$.013	0.012 $\pm$.013	0.019 $\pm$.018	0.029 $\pm$.031
<i>a — predicted mean over GT-active pixels</i>							
slot 0	0.84	0.92 $\pm$.03	0.76 $\pm$.02	0.77 $\pm$.02	0.77 $\pm$.01	0.75 $\pm$.01	0.76 $\pm$.01
slot 1	1.67	1.79 $\pm$.02	1.59 $\pm$.05	1.61 $\pm$.04	1.70 $\pm$.05	1.51 $\pm$.04	1.43 $\pm$.07
slot 2	1.40	1.68 $\pm$.08	1.50 $\pm$.06	1.51 $\pm$.04	1.70 $\pm$.11	1.38 $\pm$.08	1.41 $\pm$.09
slot 3	0.96	1.08 $\pm$.15	1.09 $\pm$.09	1.10 $\pm$.05	1.10 $\pm$.17	0.84 $\pm$.10	0.88 $\pm$.12
slot 4	0.65	0.59 $\pm$.08	0.45 $\pm$.07	0.49 $\pm$.04	0.47 $\pm$.10	0.29 $\pm$.01	0.25 $\pm$.06
<i>a — predicted mean over GT-inactive pixels (leakage)</i>							
slot 1	0.00	0.78 $\pm$.58	0.43 $\pm$.50	0.44 $\pm$.50	0.86 $\pm$.71	0.66 $\pm$.38	0.32 $\pm$.35
slot 2	0.00	1.16 $\pm$.77	0.87 $\pm$.67	0.98 $\pm$.69	0.90 $\pm$.62	0.79 $\pm$.46	0.69 $\pm$.42
slot 3	0.00	0.86 $\pm$.59	0.70 $\pm$.67	0.73 $\pm$.62	0.52 $\pm$.66	0.51 $\pm$.32	0.59 $\pm$.39
slot 4	0.00	0.58 $\pm$.60	0.58 $\pm$.62	0.57 $\pm$.61	0.42 $\pm$.55	0.34 $\pm$.48	0.43 $\pm$.45
<i>distribution width — std over GT-active pixels</i>							
a — slot 0	1.59	1.48 $\pm$.04	1.39 $\pm$.05	1.39 $\pm$.03	1.41 $\pm$.03	1.36 $\pm$.04	1.36 $\pm$.04
μ — slot 0	3.02	2.52 $\pm$.03	2.42 $\pm$.06	2.44 $\pm$.05	2.43 $\pm$.06	2.53 $\pm$.05	2.46 $\pm$.07
σ — slot 0	1.72	1.06 $\pm$.04	1.05 $\pm$.04	1.08 $\pm$.04	1.06 $\pm$.06	1.16 $\pm$.03	1.16 $\pm$.06
a — slot 1	1.60	1.40 $\pm$.03	1.35 $\pm$.02	1.36 $\pm$.03	1.44 $\pm$.04	1.42 $\pm$.03	1.39 $\pm$.06
μ — slot 1	11.08	8.85 $\pm$.26	8.58 $\pm$.18	8.59 $\pm$.32	8.81 $\pm$.27	8.93 $\pm$.28	8.76 $\pm$.20
σ — slot 1	2.30	1.44 $\pm$.04	1.62 $\pm$.04	1.60 $\pm$.04	1.46 $\pm$.02	1.62 $\pm$.02	1.71 $\pm$.05

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=5$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = paired curve term at its weight w ; none = param-L1 alone. Five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the **GT** column, marked only where best and worst differ by more than 5%. **Green** = the wake-up: slot 0 fires on 0.756 of pixels under param-L1 alone and on 0.948 with the trace-weight cosine term (0.96–0.97 at heavier w , GT 1.000); slot-1 activity climbs toward the GT (0.177 $\rightarrow$ 0.218–0.228 vs 0.239) and the slot-0 amplitude bias clears (0.92 $\rightarrow$ 0.75–0.77 vs GT 0.84). Deep-slot activity stays under the GT in every arm — the wake-up is a slot-0/1 story. Width rows: std of the prediction over GT-active pixels vs the GT's own std, shown for slots 0–1 (tail slots in the report). μ , σ in elevation units; α in linear amplitude.